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End of Year Project Report *of*

First year in Computer Engineering

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Design and Implementation of A Construction Site Monitoring System

Composition of the jury

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Academic Year: 2025-2026

Dedications

“We dedicate this work

*To our esteemed parents, for their endless sacrifices, prayers, and
unwavering support throughout our academic journey.*

*To our professors at the National Higher School of Engineers of Tunis,
who have shared their knowledge and passion for engineering with us.*

*To everyone who contributed, directly or indirectly, to the successful
completion of this project.”*

Acknowledgments

We would like to express our sincere gratitude to our supervisor, Ms. Ines Bayoudh Saadi, for her continuous support, technical guidance, and the valuable time she dedicated to our project. Her expertise was instrumental in helping us navigate the complexities of the work.

We also wish to thank the National Higher School of Engineers of Tunis and the Computer Science Department for providing the academic framework and resources necessary for the realization of this project.

Furthermore, we extend our appreciation to the members of the jury for their time, consideration, and for the upcoming evaluation of this work.

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General Introduction

The construction and civil engineering industry is characterized by the generation of a massive amount of technical, contractual, and financial data. In a sector where precision is paramount, project success depends fundamentally on the rigorous organization of information flows to optimize deadlines and ensure profitability. Today, efficient document management and real-time sharing of progress reports are major performance drivers for companies in this field.

However, many organizations struggle to centralize and secure essential documents such as blueprints, quotes, and commitment acts. Manual or fragmented monitoring of works often leads to costly errors, particularly during the preparation of work progress statements and periodic billing. Without an accurate measurement of quantities actually achieved compared to budget forecasts, the financial and technical management of a construction site becomes complex and risky.

To address these challenges, implementing a computerized "Construction Site Monitoring System" stands out as the ideal solution. This system allows for the centralization of client and project data while automating the tracking of work items and cumulative quantities. The goal is to provide an intuitive web interface backed by a robust relational database to ensure information consistency and facilitate decision-making.

It is within this framework that our Year-End Project is conducted. Our report is structured as follows:

- **Chapter 1** presents the general context of the construction industry to be modeled and the detailed requirements analysis.
- **Chapter 2** section is dedicated to the data modeling process, featuring the data dictionary as well as the conceptual and logical models.
- **Chapter 3** is where we describe the technical environment, including the choice of tools and technologies, and present the physical data model along with the SQL scripts used to create the database schema.
- **Chapter 4** is dedicated to the formulation of specific SQL queries and the presentation of their execution results.
- **Chapter 5** finally outlines the general architecture of the web application and illustrates the most relevant user interfaces.

Chapter 1. Preliminary study

Introduction

This first chapter represents the immersion phase of our project. We first present the general context of the construction sector and the host organization. Next, we provide a detailed analysis of the existing system to identify the issues related to manual construction site management. Finally, this chapter defines the functional objectives expected for the design of our new monitoring system.

1. Project Context

During a construction project, several common problems can arise and disrupt the progress of work. Delays are frequent, often caused by supply chain issues or unfavorable weather conditions. Coordination between various stakeholders can also be a challenge, leading to errors or a lack of organization. Insufficient tracking of work progress makes it difficult to detect anomalies quickly. Furthermore, budget overruns are often the result of poor resource management. Finally, the absence of a centralized system to manage information and documents complicates communication and site monitoring.

The project consists of managing a construction site using an information system designed to address these common issues encountered during execution. This system centralizes site-related information, tracks task progress, and improves coordination between the various stakeholders. It provides better visibility into the status of the work, facilitates resource management, and contributes to the reduction of delays and errors. Through computerized management, managers can quickly access the necessary data and make more effective decisions to ensure the smooth running of the project.

2. Description of the Universe to be Modeled

The universe to be modeled concerns the information system of a construction and civil engineering (BTP) firm that manages multiple construction site projects simultaneously. The firm operates in a highly structured environment where precision, traceability, and financial control are essential to the success of each project.

The firm works with a set of Clients, each identified by their full name and address. A client may commission one or several projects from the firm, but each project is always associated with a single client who acts as its sole commissioner.

Each Project is characterized by a name, a total budget, a start date, and an expected end date. A project is contractually structured by a Specification Document (referred to as SD), which defines the technical and financial scope of the work to be carried out. This document is identified by a version number and a creation date, and it serves as the contractual reference for all work items associated with the project.

The Specification Document is composed of one or several Work Packages, each representing a specific construction task such as earthworks, concrete structures, finishing works, or

technical installations. Each work package is described by its name, the initially planned quantity, the unit of measurement (such as m², m³, hours, or lumpsum), and the unit price agreed upon in the contract.

In order to monitor the financial and technical progress of each project, the firm periodically establishes Situations of Works, also known as progress statements. Each situation covers a defined period with a start date and an end date, and it belongs to one specific project. For each situation, the firm records the Achieved Quantity of every relevant work package, allowing a cumulative tracking of the work actually completed on site compared to what was initially planned.

Finally, the firm centralizes all essential Documents related to each project, including contracts, quotations, order validations, self-control sheets, and plans. Each document is identified by its type, its upload date, a URL pointing to its physical location, and an authorization level that controls who can access it within the organization.

This universe therefore revolves around six core entities — Client, Project, Specification Document, Work Package, Situation, and Document — linked together by well-defined relationships that reflect the real operational and contractual logic of a construction site management firm.

3. Critique of the Existing System

The construction industry (AEC - Architecture, Engineering, and Construction) relies on the complex management of technical, financial, and documentary data. Currently, most construction companies use traditional methods to manage information, such as office files (Excel, Word, PDF) or paper archives. This approach has several limitations:

- **Lack of speed and efficiency:** The large volume of information to manage slows down work and increases the likelihood of errors.
- **Data dispersion:** Important documents (specifications, quotes, blueprints, meeting minutes) are often scattered, making them difficult to access.
- **Inadequate progress tracking:** Progress updates and status invoices are generally handled manually, which is time-consuming and unreliable.
- **Negative financial impact:** Delays and errors in data management have additional financial consequences on project profitability.
- **Lack of centralization and security:** Documents are not always protected or version-controlled, which weakens the traceability and credibility of information.
- **Absence of intelligent decision support:** Traditional systems provide no automated analysis or guidance, leaving managers without real-time insights to support their decisions on site.

4. Requested Work

With the goal of digitalization and improving data management in the construction industry, we propose the creation and implementation of a relational database accessible through a web interface, enhanced with an integrated AI assistant to support decision-making.

This solution ensures:

- Organization of data, information, and documents: All project-related information (clients, specifications, work items, progress reports, contractual documents) will be centralized in a single system.
- Easy and fast access to data: The web interface will allow for real-time information consultation from any workstation within the firm.
- Simplified data manipulation: The ability to track work progress, update data seamlessly, and monitor the financial health of each project through automatically computed indicators.
- Intelligent decision support: An integrated AI chatbot powered by a local Ollama language model provides users with instant guidance on construction management concepts, financial interpretation, and system usage — without requiring any external connection, ensuring full data confidentiality.

Conclusion

In conclusion, the preliminary study has allowed us to precisely identify the gaps in the current system and lay the groundwork for our analysis. This step was essential to define the scope of our IT solution. Now that the requirements are identified, we will move on to chapter 2 to begin the detailed study and the conceptual modeling of the database.

Chapter 2. Detailed Study

Introduction

The second chapter includes the refined data dictionary, the business rules, as well as the Conceptual Data Model (CDM) and the Logical Data Model (LDM).

1. Data Collection and Analysis

In order to create a raw data dictionary, we conducted a descriptive analysis and identified a dataset from which we removed synonyms, polysemous terms, and calculated data, keeping only the essential data elements.

2. Data Dictionary

The refined data dictionary is described in Table 1.

Code	Designation	Data Type & length	Constraints
Project_id	Identity of the project	Number (8)	Identity of the entity Project
Project_name	Name of the project	Variable characters (20)	
Budget	Budget of the project	Decimal (10,3)	Must be > 0
Project_start_date	Start date of the project	Date	
Project_expected_end_date	Expected End date of the project	Date	
Client_id	Identity of the client	Number (8)	Identity of the entity Client
First_name	First Name of the client	Variable characters (20)	
Last_name	Last Name of the client	Variable characters (20)	
Address	Address of the client	Variable characters (30)	
Document_id	Identity of the	Number (8)	Identity of the entity

	Document		Document
Document_type	Type of Document	Variable characters (20)	$\text{Doc_Type} \in \{\text{'Contract'}, \text{'Quotation'}, \text{'Order Validation'}, \text{'Self-Control Sheet'}, \text{'Approval'}, \text{'Plan'}\}$
Document_url	Link to the Document	Variable characters (255)	$\text{Document_url} \in \{u \mid u \text{ starts with } \text{'/doc/' or 'https://'}\}$
Upload_date	The upload date of the document	Date	
Authorization_level	The access level required to view the document	Variable characters (20)	$\text{Authorization_Level} \in \{\text{'Public'}, \text{'Internal'}, \text{'Confidential'}\}$
SD_id	Identity of the the specifications	Number (8)	Identity of the entity specifications
Creation_date	Creation date of the specifications	Date	
Version	Version of the specifications	Variable characters (20)	$\text{Version} \in \{ \text{'v'} + n \mid n \in \mathbb{N}^* \}$
WorkP_id	Identity of the work Package	Number (8)	Identity of the entity Work Package
WorkP_name	Name of the work Package	Variable characters (20)	
Expected_quantity	The initially planned quantity	Number (8)	
Measurment_unit	The unit of measurement	Variable characters (10)	
Unit_price	The unit price	Decimal(10,3)	
Situation_id	Identity of the situation	Number (8)	Identity of the entity Situation
Comments	Additional remarks	Variable Characters (100)	
Start_date	Start date of the situation	Date	
End_date	End date of the	Date	

	situation		
Achieved_quantity	Quantity Actually Achieved	Number (8)	

Table 1: Data dictionary

3. Management Rules

In this section, we describe the rules according to their types.

3.1. Rules of type “Definition”

In this section, we describe the definition-based rules:

- A **Project** is described by its ID, name, budget, start date, and expected end date.
- A **Client** is described by its ID, first name, last name, and address.
- A **Document** is described by its ID, type, URL, upload date and authorization level.
- A **Specification Document** is described by its ID, creation date and version.
- A **Work Package** is described by its ID, name, expected quantity, measurement unit, and unit price.
- A **Situation** is described by its ID, start date, end date, and comments.

3.2. Rules of Type “Fact”

In this section, we describe the fact-based rules:

- A **Project** centralizes one or many documents.
- A **Document** is centralized by one and only one project.
- A **Project** is structured by one and only one Specification Document
- A **Specification Document** structures none or one Project
- A **Specification Document** contains one or many work Packages.
- A **Work Package** is contained in one and only one specifications document.
- A **Client** orders one or many Projects.
- A **Project** is ordered by one and only one Client.
- A **Project** establishes one or many situations.
- A **Situation** is established by one and only one project.
- A **Work Package** achieves a certain **Achieved Quantity** in a given **Situation**.
- A **Situation** has an **Achieved Quantity** corresponding to at least a certain **Work Packages**.

3.3. Rules of Type “Constraint”

In this section, we describe the constraint-type rules:

- Each entity must have a unique and non-null identifier .
- The end date of a project must be greater than or equal to its start date.
- A budget must be positive.
- An expected quantity must be non-negative.
- A unit price must be positive.
- The document type must belong to the set { 'Contract', 'Quotation', 'Order Validation', 'Self-Control Sheet', 'Approval', 'Plan' }.

3.4. Rules of Type “Need”

In this section, we describe the need-type rules:

- A **Project** must have at least one Specifications document to be valid.
- A **Specification Document** must contain at least one work Package.

3.5. Rules of Type “Formula”

In this section, we describe the rules of the Formula type:

- The **total cost of a work Package** is calculated as:

$$\text{Total Cost} = \text{Expected Quantity} \times \text{Unit Price}$$

- The **duration of a situation** is calculated as :

$$\text{Duration} = \text{End_date} - \text{Start_date}$$

4. *Conceptual Data Modeling*

The Conceptual Data Model is illustrated in Figure 1.

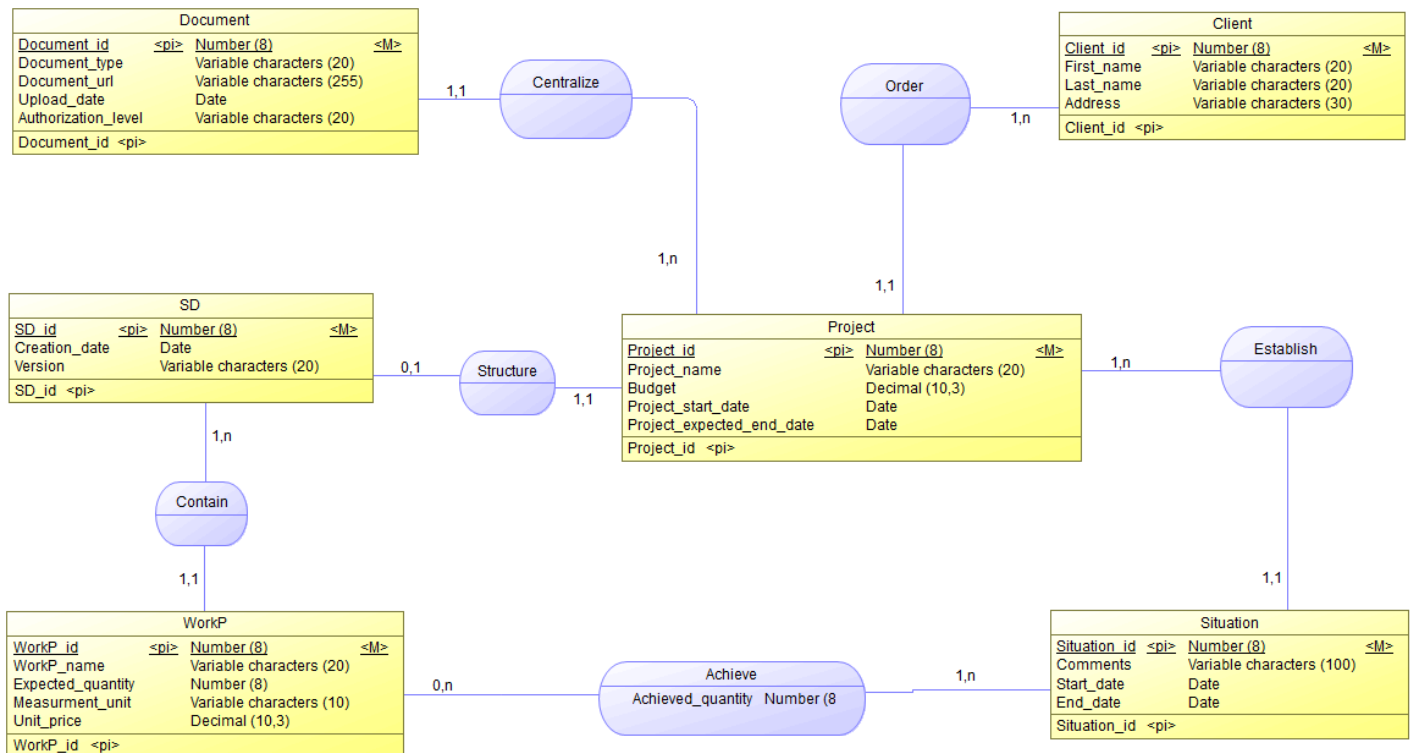


Figure 1: Conceptual Data Model

4.1. Description of the entities

The entities are described in Table 2.

Entity	Description														
Project <table border="1"> <thead> <tr> <th colspan="2">Project</th> </tr> </thead> <tbody> <tr> <td>Project_id</td><td><pi> Number (8) <M></td></tr> <tr> <td>Project_name</td><td>Variable characters (20)</td></tr> <tr> <td>Budget</td><td>Decimal (10,3)</td></tr> <tr> <td>Project_start_date</td><td>Date</td></tr> <tr> <td>Project_expected_end_date</td><td>Date</td></tr> <tr> <td>Project_id</td><td><pi></td></tr> </tbody> </table> Project_id → Project_Name, Budget, Project_start_date, Project_expected_end_date	Project		Project_id	<pi> Number (8) <M>	Project_name	Variable characters (20)	Budget	Decimal (10,3)	Project_start_date	Date	Project_expected_end_date	Date	Project_id	<pi>	The entity “Project” is described by the following attributes: Project_id: Project’s identity. Project_name: Project’s name. Budget: Project’s budget. Project_start_date: Project’s start date. Project_expected_end_date: Project’s expected end date. It is identified by the unique number assigned to the project.
Project															
Project_id	<pi> Number (8) <M>														
Project_name	Variable characters (20)														
Budget	Decimal (10,3)														
Project_start_date	Date														
Project_expected_end_date	Date														
Project_id	<pi>														

Client <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th colspan="2" style="text-align: center;">Client</th> </tr> </thead> <tbody> <tr> <td><u>Client_id</u></td> <td style="text-align: right;"><pi> Number (8) <M></td> </tr> <tr> <td>First_name</td> <td style="text-align: right;">Variable characters (20)</td> </tr> <tr> <td>Last_name</td> <td style="text-align: right;">Variable characters (20)</td> </tr> <tr> <td>Address</td> <td style="text-align: right;">Variable characters (30)</td> </tr> <tr> <td colspan="2"><u>Client_id</u> <pi></td> </tr> </tbody> </table> Client_id → First_name, Last_name, Address	Client		<u>Client_id</u>	<pi> Number (8) <M>	First_name	Variable characters (20)	Last_name	Variable characters (20)	Address	Variable characters (30)	<u>Client_id</u> <pi>		The entity “Client” is described by the following attributes: Client_id: Client’s identity. First_name: Client’s name. Last_name: Client’s last name. Address: Client’s address. It is identified by the unique number assigned to the client.		
Client															
<u>Client_id</u>	<pi> Number (8) <M>														
First_name	Variable characters (20)														
Last_name	Variable characters (20)														
Address	Variable characters (30)														
<u>Client_id</u> <pi>															
Document <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th colspan="2" style="text-align: center;">Document</th> </tr> </thead> <tbody> <tr> <td><u>Document_id</u></td> <td style="text-align: right;"><pi> Number (8) <M></td> </tr> <tr> <td>Document_type</td> <td style="text-align: right;">Variable characters (20)</td> </tr> <tr> <td>Document_url</td> <td style="text-align: right;">Variable characters (255)</td> </tr> <tr> <td>Upload_date</td> <td style="text-align: right;">Date</td> </tr> <tr> <td>Authorization_level</td> <td style="text-align: right;">Variable characters (20)</td> </tr> <tr> <td colspan="2"><u>Document_id</u> <pi></td> </tr> </tbody> </table> ID_Document → Document_type, Document_url, Upload_date, Authorization_level	Document		<u>Document_id</u>	<pi> Number (8) <M>	Document_type	Variable characters (20)	Document_url	Variable characters (255)	Upload_date	Date	Authorization_level	Variable characters (20)	<u>Document_id</u> <pi>		The entity “Document” is described by the following attributes: Document_id: Document’s identity. Document_type: Document’s type. Document_url: Document’s file source. Upload_date: Document’s date of first upload. Authorization_level: Document’s authorization level. It is identified by the unique number assigned to the document.
Document															
<u>Document_id</u>	<pi> Number (8) <M>														
Document_type	Variable characters (20)														
Document_url	Variable characters (255)														
Upload_date	Date														
Authorization_level	Variable characters (20)														
<u>Document_id</u> <pi>															
WorkP <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th colspan="2" style="text-align: center;">WorkP</th> </tr> </thead> <tbody> <tr> <td><u>WorkP_id</u></td> <td style="text-align: right;"><pi> Number (8) <M></td> </tr> <tr> <td>WorkP_name</td> <td style="text-align: right;">Variable characters (20)</td> </tr> <tr> <td>Expected_quantity</td> <td style="text-align: right;">Number (8)</td> </tr> <tr> <td>Measurment_unit</td> <td style="text-align: right;">Variable characters (10)</td> </tr> <tr> <td>Unit_price</td> <td style="text-align: right;">Decimal (10,3)</td> </tr> <tr> <td colspan="2"><u>WorkP_id</u> <pi></td> </tr> </tbody> </table> WorkP_id → WorkP_name, Expected_quantity, Measurement_unit, Unite_price	WorkP		<u>WorkP_id</u>	<pi> Number (8) <M>	WorkP_name	Variable characters (20)	Expected_quantity	Number (8)	Measurment_unit	Variable characters (10)	Unit_price	Decimal (10,3)	<u>WorkP_id</u> <pi>		The entity “WorkP” is described by the following attributes: WorkP_id: Work Package’s identity. WorkP_name: Work Package’s name. Expected_quantity: Package’s expected quantity of work. Measurement_unit: Work Package’s unit of measure. Unit_price: Work Package’s unit price. It is identified by the unique number assigned to the work Package.
WorkP															
<u>WorkP_id</u>	<pi> Number (8) <M>														
WorkP_name	Variable characters (20)														
Expected_quantity	Number (8)														
Measurment_unit	Variable characters (10)														
Unit_price	Decimal (10,3)														
<u>WorkP_id</u> <pi>															
Situation <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th colspan="2" style="text-align: center;">Situation</th> </tr> </thead> <tbody> <tr> <td><u>Situation_id</u></td> <td style="text-align: right;"><pi> Number (8) <M></td> </tr> <tr> <td>Comments</td> <td style="text-align: right;">Variable characters (100)</td> </tr> <tr> <td>Start_date</td> <td style="text-align: right;">Date</td> </tr> <tr> <td>End_date</td> <td style="text-align: right;">Date</td> </tr> <tr> <td colspan="2"><u>Situation_id</u> <pi></td> </tr> </tbody> </table> Situation_id → Comments, Start_date, End_date	Situation		<u>Situation_id</u>	<pi> Number (8) <M>	Comments	Variable characters (100)	Start_date	Date	End_date	Date	<u>Situation_id</u> <pi>		The entity “Situation” is described by the following attributes: Situation_id: Situation’s identity. Comments: Situation’s comments. Start_date: Situation’s start date. End_date: Situation’s end date. It is identified by the unique number assigned to the work situation.		
Situation															
<u>Situation_id</u>	<pi> Number (8) <M>														
Comments	Variable characters (100)														
Start_date	Date														
End_date	Date														
<u>Situation_id</u> <pi>															

SD	
SD	
SD_id	<u><pi> Number (8) <M></u>
Creation_date	Date
Version	Variable characters (20)
SD_id	<pi>

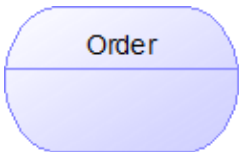
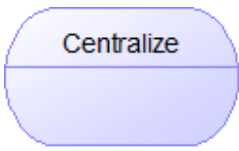
SD_id → Creation_date, Version

The entity “SD” is described by the following attributes:
SD_id: Specification document’s identity.
Creation_date: Specification document’s date of creation
Version: Specification document’s version
It is identified by the unique number assigned to the Specification document.

Table 2: Entity descriptions

4.2. Description of relations and cardinalities

Relations represent the existing links between the entities. The table 3 provides a detailed description of the relations.

Relation	Entity	Cardinality	Description
	Project - Client	1,1 - 1,n	The relation "Order" has no inherent attribute, its dimension is 2. It links the project to the client. It is characterized by the pair of cardinality (1,1)-(1,n) since a project is ordered by only one client and a client orders one or several projects.
	Document - Project	1,1 - 1,n	The relation "Centralize" has no inherent attribute, its dimension is 2. It links the document to the project. It is characterized by the pair of cardinality (1,1)-(1,n) since a document is centralized by only one project and a project centralizes one or several documents.

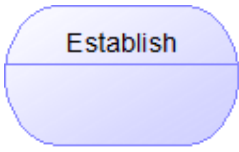
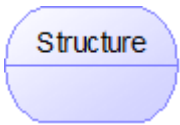
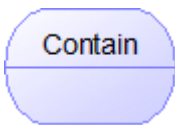
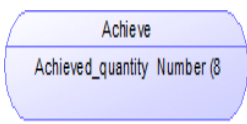
	Project Situation	- 1,n - 1,1	The relation " Establish " has no inherent attribute, its dimension is 2. It links the project to the situation. It is characterized by the pair of cardinality (1,n)-(1,1) since a situation is established by a project to happen and a project can establish one or several situations.
	Project - SD	1,1 - 0,1	The relation " Structure " has no inherent attribute, its dimension is 2. It links the project to the specifications . It is characterized by the pair of cardinality (1,1)-(0,1) since a project is structured by one and only specifications and a specification document can structure none or one project.
	SD - WorkP	1,n - 1,1	The relation " Contain " has no inherent attribute, its dimension is 2. It links the Specifications and the work Package. It is characterized by the pair of cardinality (1,n)-(1,1) since a work Package is contained in only one specification Document and a specification Document can contain one or several work Packages.
	Situation WorkP	- 1,n - 0,n	The relation " Achieve " has an inherent attribute named " Achieved_quantity ", its dimension is 2. It links the situation and the work Package. It is characterized by the pair of cardinality (1,n)-(0,n) since a situation describes the achievement of one or many work packages and a work Package is achieved in none or many situations.

Table 3: Relationship descriptions

4.3. Verification of Normalization Rules

Our objective is to develop a verified and normalized model in Third Normal Form (3NF). To this end, we will define each of the normal forms and apply them to our model.

- **First Normal Form (1NF):** All attributes of the entity must contain an atomic value, meaning they are non-multiple and non-decomposable.
- **Second Normal Form (2NF):** An entity satisfies 2NF if it satisfies 1NF and if the attributes of each entity depend fully on the primary key.
- **Third Normal Form (3NF):** The third normal form is satisfied if 2NF is met and when there is no dependency between non-key attributes.

As an example, we will consider the Client entity for the verification of normalization rules.

- Verification of the **First Normal Form**: From the Client entity, for example, all attributes contain an atomic value, meaning they are non-multiple and non-decomposable.
- Verification of the **Second Normal Form**: We will consider the same entity (Client) which is in 1NF and the following functional dependencies:
 - ◆ Client_id → First_name
 - ◆ Client_id → Last_name
 - ◆ Client_id → Address

We can deduce that all attributes depend fully on the identifier; therefore, it is in 2NF.

- Verification of the **Third Normal Form**: Since the *Client* entity satisfies the Second Normal Form (2NF), and there are no functional dependencies between non-key attributes—for example, the client's last name does not determine the first name and vice versa—the entity is therefore in 3NF.

In Table 4, we validated the normalization rules for all attributes.

Entity	1NF	2NF	3NF
Project	✓	✓	✓
Client	✓	✓	✓
Document	✓	✓	✓
SD	✓	✓	✓
WorkP	✓	✓	✓
Situation	✓	✓	✓

Table 4: Normalization rule verification

5. Logical Data Modeling

The Logical Data Model (LDM) is an intermediate model used to move from the E/R model, which is a semantic model, to a physical representation of the data.

We generated the LDM from the CDM through the steps described below, which are illustrated in Figure 2.

1. Click on **Tools /Generate Logical Data Model..**. A window appears with the title "*LDM Generation Options*".
2. Enter a name for the Logical Data Model, then click **OK**.

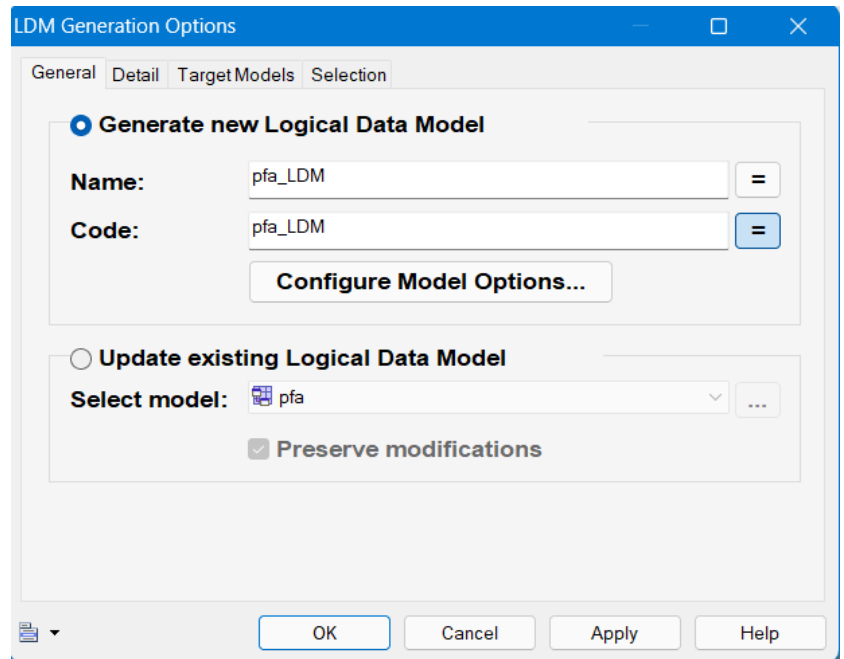
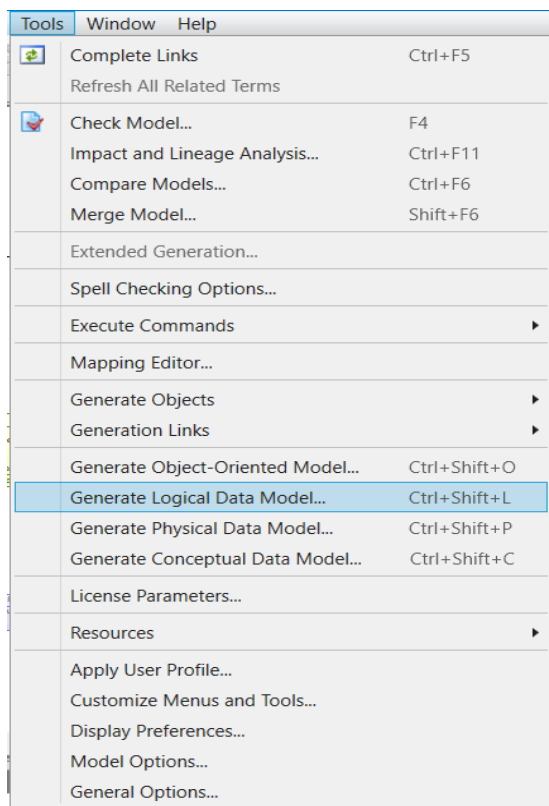


Figure 2 : Generating LDM

The Logical Data Model has been successfully created. It is illustrated in Figure 3.

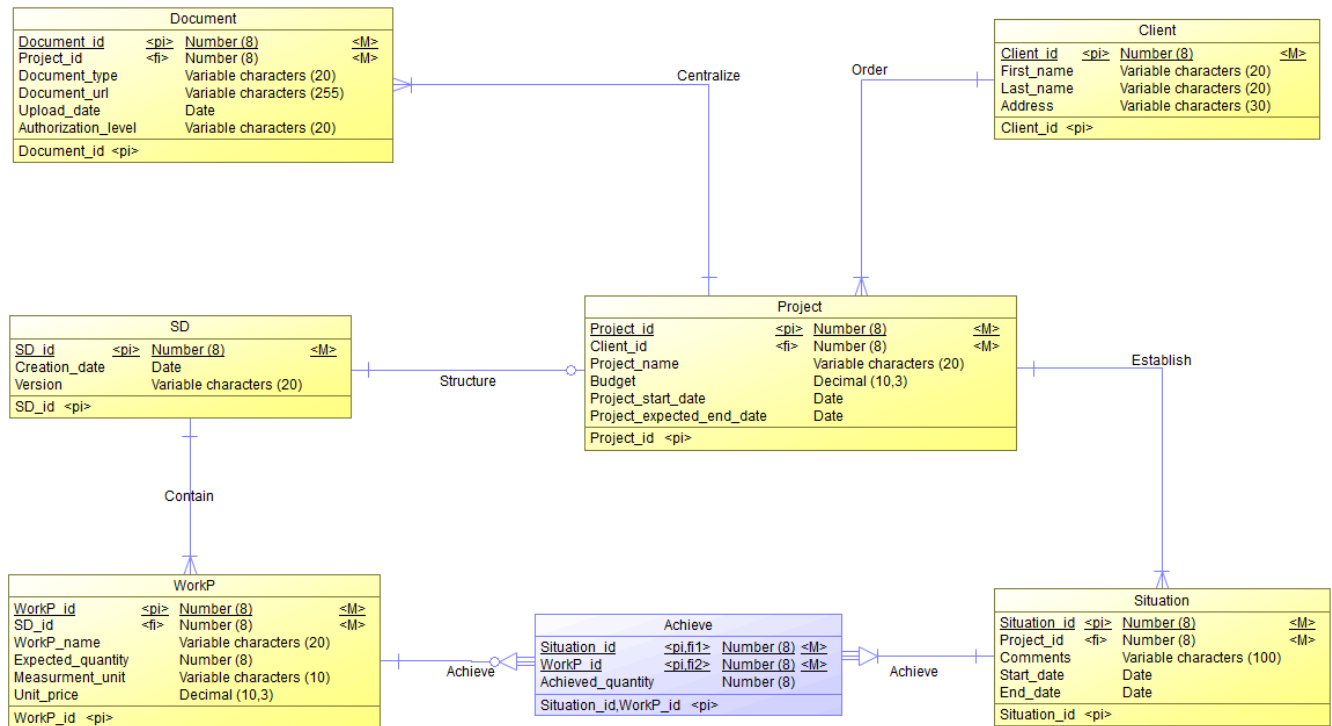


Figure 3 : Logical Data Model

Conclusion

After determining a refined data dictionary and the management rules governing our environment, representing the problem through a conceptual data model while describing the entities, relationships, and cardinalities, verifying that the model is normalized, and generating a logical data model, we will in the next chapter focus on the technical study of our project.

Chapter 3. Technical Study

Introduction

In this chapter, we present the technical environment used for the realization of our project. We first describe the hardware and software environments adopted during the development process. Then, we introduce the Physical Data Model (PDM) and explain the steps followed for database creation and initialization with the required data.

1. Hardware Environment

This project was carried out using a personal computer with the following specifications:

PC Characteristics	Description
Processor	Intel Core i7
RAM	24 GB
Operating System	Windows 11 Home Single Language

Table 5: Hardware environment

2. Software Environment

The following software tools were used throughout the development and implementation of this project:

Software	Logo	Description
SAP PowerDesigner 16.7		SAP PowerDesigner [1] (SAP SE, n.d.) (Version 16.7 SP05, 64-bit) is a data modeling and enterprise architecture tool used for designing information systems and databases. It was used to create and transform the Conceptual Data Model (CDM), Logical Data Model (LDM), and Physical Data Model (PDM), as well as to generate the SQL scripts for database creation.

Oracle Database 11g		Oracle Database [2] is a multi-model database management system developed by Oracle Corporation. It was used to create, store, and manage the relational database of the application.
Oracle SQL Developer		Oracle SQL Developer [3] is a free integrated development environment provided by Oracle that allows users to execute SQL queries, manage database objects, and administer Oracle databases through a graphical interface.
Visual Studio Code		Visual Studio Code [4] is a cross-platform code editor developed by Microsoft. It supports multiple programming languages such as HTML, CSS, JavaScript, PHP, and others, and is used for application development and source code editing.
XAMPP		XAMPP [5] is a free and open-source cross-platform web server solution stack. It is a powerful and easy-to-use tool that allows for the rapid deployment of a local development environment. In the context of our project, XAMPP served as our primary local server environment, enabling us to manage the database and host our dashboard's web interface simultaneously.

Table 6: Software tools and technologies

3. Physical Data Model

A **Physical Data Model (PDM)** is a representation that describes the actual implementation of a database by defining tables, columns, indexes, keys, and constraints according to a specific Database Management System (DBMS). Unlike the Conceptual Data Model (CDM) and the Logical Data Model (LDM), which focus on business concepts and logical structure,

the PDM provides a concrete view of how data is physically stored and organized within the database system.

The Physical Data Model allows the analysis and definition of database objects such as tables, relationships, constraints, and indexes required for database implementation.

The steps followed to create the Physical Data Model using **SAP PowerDesigner** are illustrated in Figure 4 and include:

1. Selecting **Tools** → **Generate Physical Data Model** from the menu.
2. A dialog window titled *PDM Generation Options* appears.
3. Choosing the target DBMS (Oracle Database).
4. Validating the generation process by clicking *OK* to automatically transform the logical model into a physical model.

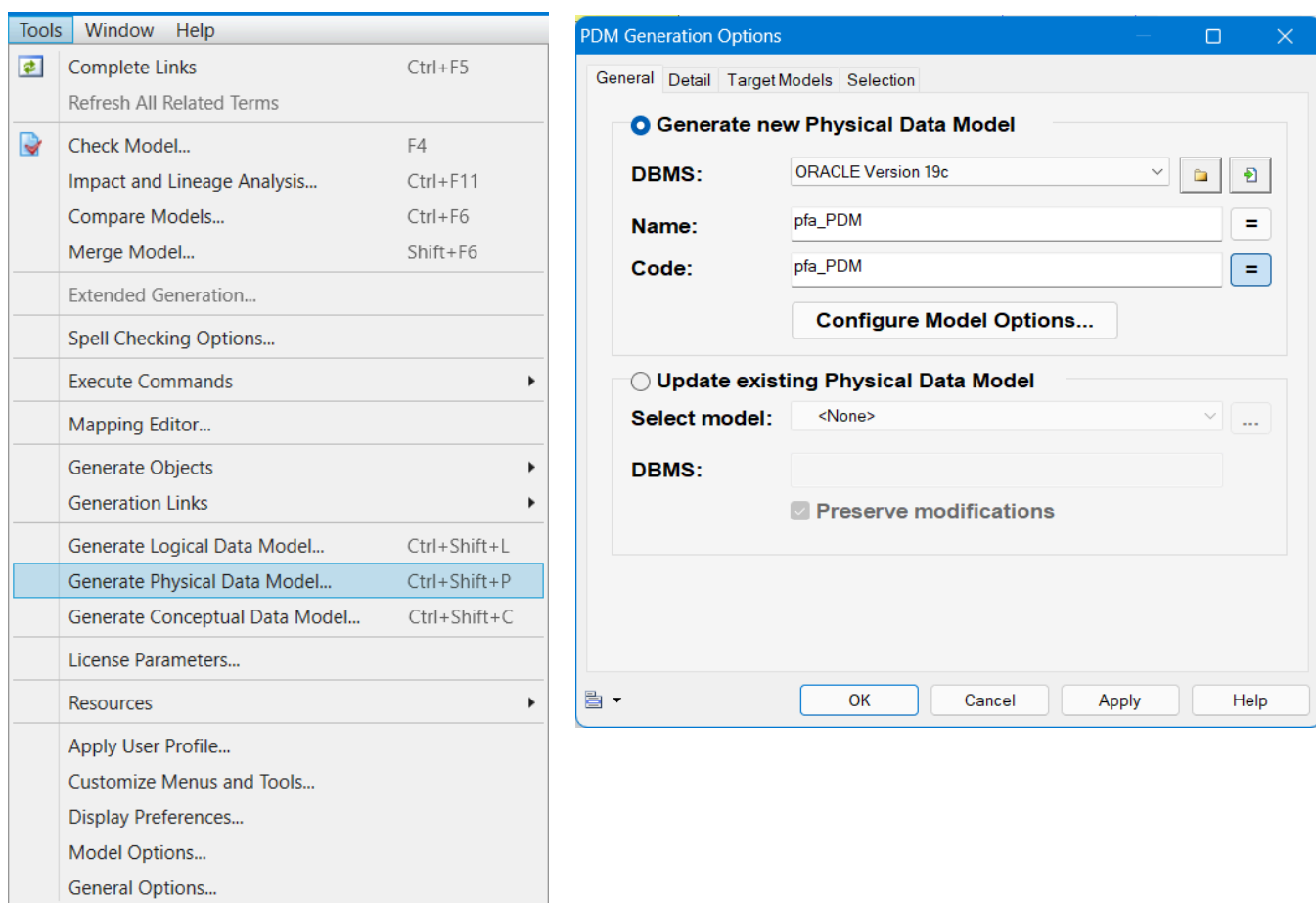


Figure 4 : Generating PDM

After completing these steps, the Physical Data Model was successfully generated and is illustrated in Figure 5.

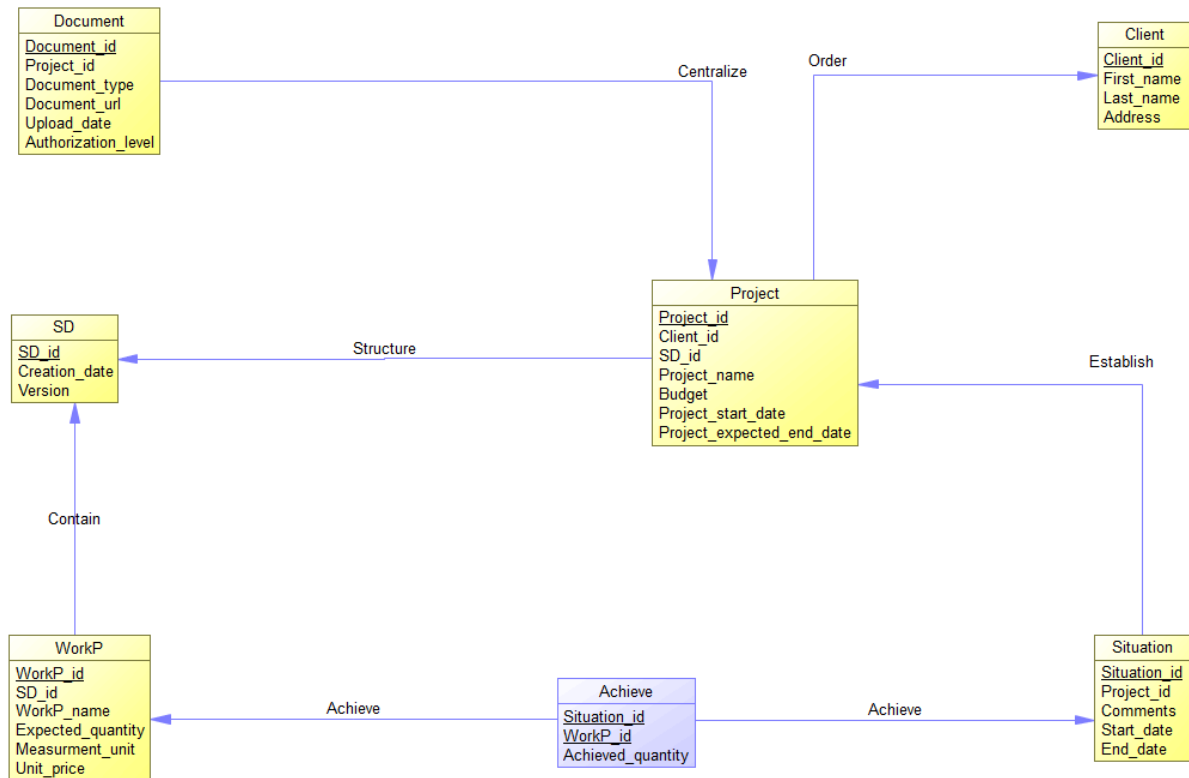


Figure 5 : Physical Data Model

4. Database Creation

In this section, we detail the steps required to generate and execute the database creation script.

4.1. Generation of the Database Creation Script

SAP PowerDesigner allows the automatic generation of a database creation script that can be executed later within the DBMS environment. It also provides the possibility to directly generate the database structure through an active database connection (Figure 6).

The database generation process was carried out as follows:

1. Selecting **Database** → **Generate Database..** to open the *Database Generation* dialog box.
2. Specifying the destination directory and the script file name in the corresponding fields.
3. Choosing the option **Script Generation** to create the SQL script instead of directly executing the database creation.
4. Confirming the configuration by clicking **OK**, which generated the SQL script containing all tables, constraints, and relationships.

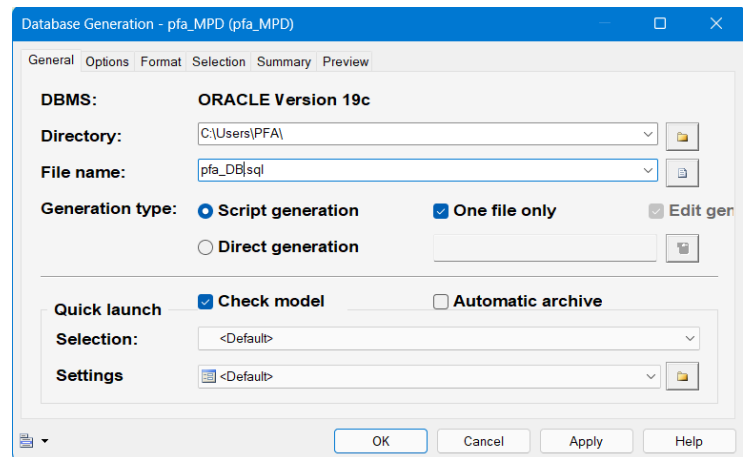
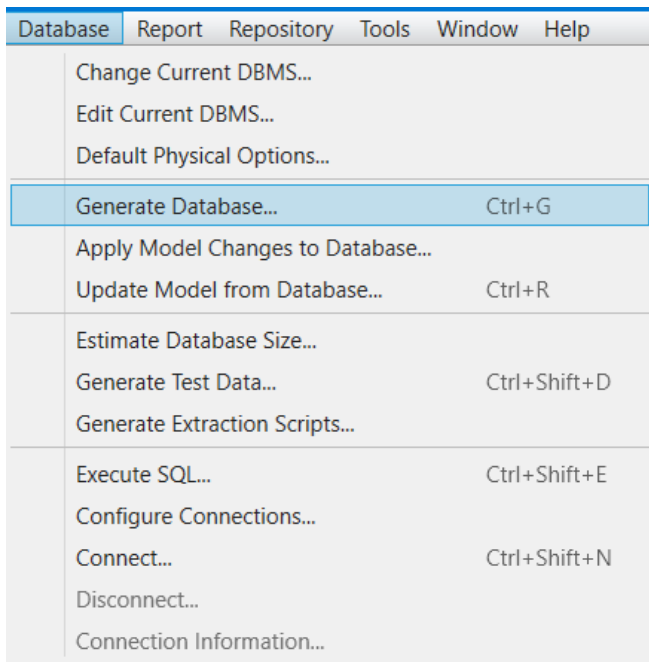


Figure 6 : Generating the database creation script

The generated script was then executed using Oracle SQL Developer to create the database structure successfully.(Figure 7)

```

/*=====*/
/* Table: "Client" */
/*=====*/
create table "Client"
(
  "Client_id"          NUMBER(8)          not null,
  "First_name"         VARCHAR2(20),
  "Last_name"          VARCHAR2(20),
  "Address"            VARCHAR2(30),
  constraint PK_CLIENT primary key ("Client_id")
);

/*=====*/
/* Table: "Document" */
/*=====*/
create table "Document"
(
  "Document_id"        NUMBER(8)          not null,
  "Project_id"         NUMBER(8)          not null,
  "Document_type"      VARCHAR2(20),
  "Document_url"       VARCHAR2(255),
  "Upload_date"        DATE,
  "Authorization_level" VARCHAR2(20),
  constraint PK_DOCUMENT primary key ("Document_id")
);

```

Figure 7: Excerpt from the database creation script

4.2. Execution of the Database Creation Script

Oracle SQL Developer provides the ability to connect to an existing database in order to visualize database objects, manage relationships between tables, and execute SQL scripts. This tool was used to run the database creation script generated previously using SAP PowerDesigner (as illustrated in Figure 8).

The steps followed to execute the script are described below:

1. Selecting “**New Connection**” from the Oracle SQL Developer interface to create a new database connection.
2. Entering the required connection information, including:
 - a. a connection name,
 - b. the username
 - c. password used to access the database
 - d. **SID**, which represents the database instance name.
3. Clicking “**Test**” to verify the connection parameters. When the status displays “**Success**”, the connection is considered valid.
4. Selecting “**Connect**” to establish the connection to the Oracle database.
5. Opening the generated SQL script file and executing it using the **Run Script** option in order to create all database tables, constraints, and relationships.

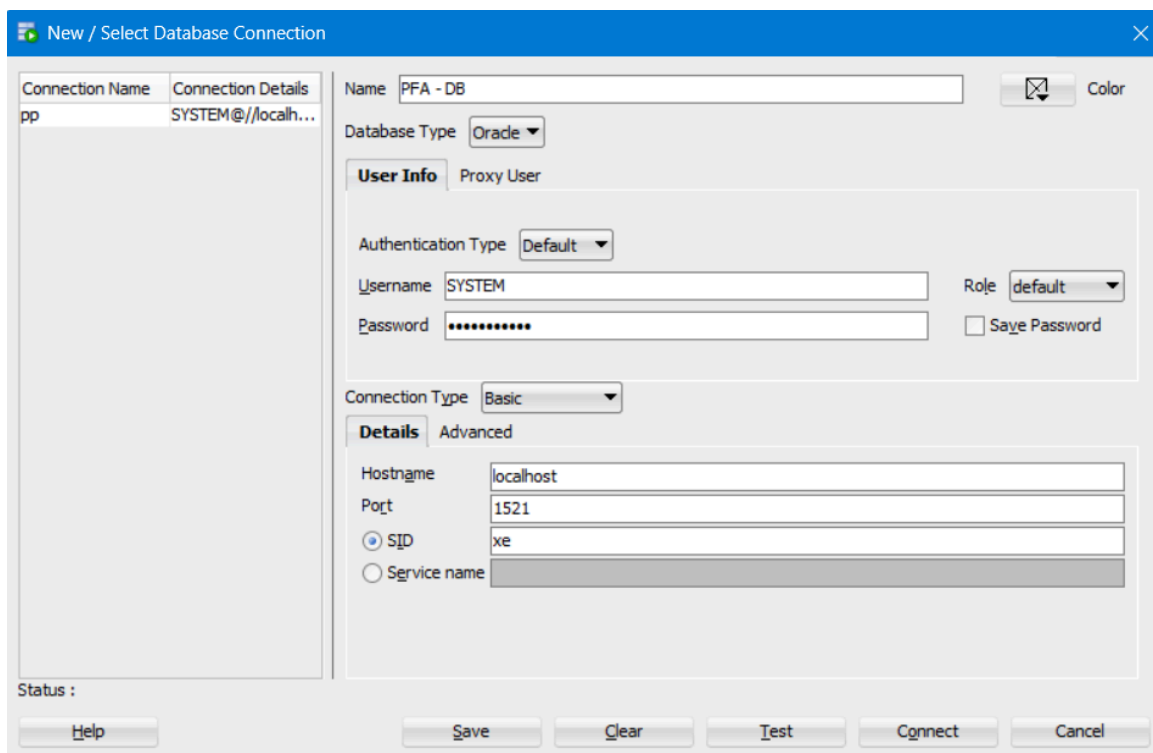


Figure 8: Database connection interface

After executing the script successfully, the database structure was created and became accessible through Oracle SQL Developer for further data manipulation and verification. This is illustrated by Figure 9.

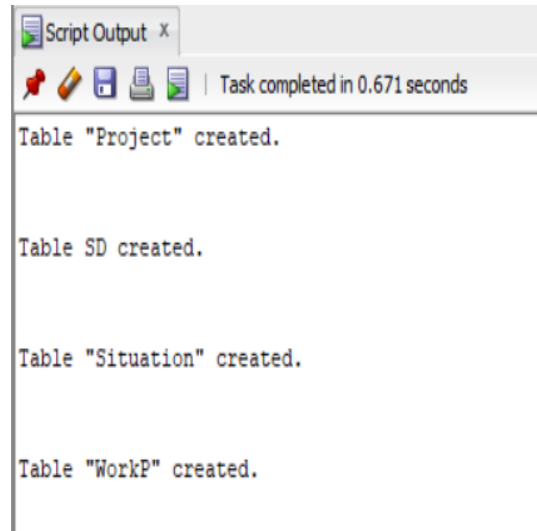


Figure 9 : Database creation Output

5. Database Population

After creating the database structure, data tuples were inserted into each table in order to test the system functionality and validate relationships between entities.

The population process followed a logical order respecting referential integrity constraints, meaning that parent tables were filled before dependent tables.

The database population scripts are presented below.

Populating the table 'Client'

```
INSERT INTO "Client" VALUES (1, 'Ahmed', 'Ben Salah', '12 Carthage Street, Tunis');
INSERT INTO "Client" VALUES (2, 'Fatma', 'Trabelsi', '45 Bourguiba Avenue, Sfax');
INSERT INTO "Client" VALUES (3, 'Karim', 'Gharbi', '8 Ibn Khaldoun Street, Sousse');
INSERT INTO "Client" VALUES (4, 'Leila', 'Chaabane', '22 Republic Street, Nabeul');
INSERT INTO "Client" VALUES (5, 'Amine', 'Missaoui', '67 Environment Avenue, Bizerte');
INSERT INTO "Client" VALUES (6, 'Sonia', 'Khelifi', '3 Tahar Haddad Street, Monastir');
INSERT INTO "Client" VALUES (7, 'Riadh', 'Jebali', '15 Roses Street, Gabes');
INSERT INTO "Client" VALUES (8, 'Nadia', 'Bouaziz', '9 Tunis Avenue, Kairouan');
INSERT INTO "Client" VALUES (9, 'Hassen', 'Dridi', '31 Farhat Hached Street, Ariana');
INSERT INTO "Client" VALUES (10, 'Slim', 'Zouari', '77 Bechir Sfar Avenue, Sfax');
```

Populating the table 'Project'

```
INSERT INTO "Project" VALUES (1, 1, 1, 'Carthage Residency', 1250000.000, TO_DATE('2023-02-01', 'YYYY-MM-DD'), TO_DATE('2024-06-30', 'YYYY-MM-DD'));
INSERT INTO "Project" VALUES (2, 2, 2, 'Sousse Beach Hotel', 1850000.000, TO_DATE('2023-07-01', 'YYYY-MM-DD'), TO_DATE('2025-06-30', 'YYYY-MM-DD'));
INSERT INTO "Project" VALUES (3, 3, 3, 'Bizerte Complex', 975000.000, TO_DATE('2024-04-01', 'YYYY-MM-DD'), TO_DATE('2026-03-31', 'YYYY-MM-DD'));
INSERT INTO "Project" VALUES (4, 4, 4, 'Ariana Tech Park', 780000.000, TO_DATE('2024-10-01', 'YYYY-MM-DD'), TO_DATE('2025-09-30', 'YYYY-MM-DD'));
INSERT INTO "Project" VALUES (5, 5, 5, 'Hammamet Clinic', 1100000.000, TO_DATE('2025-03-01', 'YYYY-MM-DD'), TO_DATE('2027-02-28', 'YYYY-MM-DD'));
INSERT INTO "Project" VALUES (6, 6, 6, 'Sfax Warehouse', 410000.000, TO_DATE('2025-09-01', 'YYYY-MM-DD'), TO_DATE('2026-12-31', 'YYYY-MM-DD'));
INSERT INTO "Project" VALUES (7, 7, 7, 'Tunis Lake Tower', 1950000.000, TO_DATE('2026-02-01', 'YYYY-MM-DD'), TO_DATE('2028-01-31', 'YYYY-MM-DD'));
INSERT INTO "Project" VALUES (8, 8, 8, 'Kairouan School', 310000.000, TO_DATE('2026-03-15', 'YYYY-MM-DD'), TO_DATE('2027-06-30', 'YYYY-MM-DD'));
INSERT INTO "Project" VALUES (9, 9, 9, 'Monastir Parking', 155000.000, TO_DATE('2026-01-20', 'YYYY-MM-DD'), TO_DATE('2027-01-19', 'YYYY-MM-DD'));
INSERT INTO "Project" VALUES (10, 10, 10, 'Nabeul North Villa', 480000.000, TO_DATE('2025-05-01', 'YYYY-MM-DD'), TO_DATE('2026-08-31', 'YYYY-MM-DD'));
INSERT INTO "Project" VALUES (11, 1, 11, 'Carthage Annex B', 390000.000, TO_DATE('2024-01-01', 'YYYY-MM-DD'), TO_DATE('2024-12-31', 'YYYY-MM-DD'));
INSERT INTO "Project" VALUES (12, 2, 12, 'Sousse Swimming Pool', 220000.000, TO_DATE('2023-06-01', 'YYYY-MM-DD'), TO_DATE('2024-05-31', 'YYYY-MM-DD'));
INSERT INTO "Project" VALUES (13, 3, 13, 'Industrial Zone Exp.', 850000.000, TO_DATE('2024-06-01', 'YYYY-MM-DD'), TO_DATE('2026-05-31', 'YYYY-MM-DD'));
INSERT INTO "Project" VALUES (14, 5, 14, 'Hammamet Apartment', 630000.000, TO_DATE('2025-01-01', 'YYYY-MM-DD'), TO_DATE('2025-12-31', 'YYYY-MM-DD'));
INSERT INTO "Project" VALUES (15, 6, 15, 'Monastir Event Hall', 180000.000, TO_DATE('2026-02-01', 'YYYY-MM-DD'), TO_DATE('2026-11-30', 'YYYY-MM-DD'));
```

Populating the table 'WorkP'

```
-- SD 1 - Carthage Residency
INSERT INTO "WorkP" VALUES (1, 1, 'Earthworks', 500, 'm3', 45.000);
INSERT INTO "WorkP" VALUES (2, 1, 'Reinforced Con.', 800, 'm3', 320.000);
INSERT INTO "WorkP" VALUES (3, 1, 'Masonry', 2400, 'm2', 85.000);
INSERT INTO "WorkP" VALUES (4, 1, 'Facade Coating', 1800, 'm2', 38.000);
-- SD 2 - Sousse Beach Hotel
INSERT INTO "WorkP" VALUES (5, 2, 'Foundations', 700, 'm3', 290.000);
INSERT INTO "WorkP" VALUES (6, 2, 'RC Structure', 1100, 'm3', 315.000);
INSERT INTO "WorkP" VALUES (7, 2, 'Alu. Joinery', 480, 'm1', 210.000);
INSERT INTO "WorkP" VALUES (8, 2, 'Interior Paint', 4500, 'm2', 18.000);
-- SD 3 - Bizerte Complex
INSERT INTO "WorkP" VALUES (9, 3, 'Earthworks', 400, 'm3', 44.000);
INSERT INTO "WorkP" VALUES (10, 3, 'Slab Concrete', 560, 'm3', 300.000);
INSERT INTO "WorkP" VALUES (11, 3, 'Roof Cover', 900, 'm2', 95.000);
-- SD 4 - Ariana Tech Park
INSERT INTO "WorkP" VALUES (12, 4, 'Earthworks', 900, 'm3', 46.000);
INSERT INTO "WorkP" VALUES (13, 4, 'Reinforced Con.', 1200, 'm3', 325.000);
INSERT INTO "WorkP" VALUES (14, 4, 'Cable Networks', 1, 'lumpsum', 72000.000);
-- SD 5 - Hammamet Clinic
INSERT INTO "WorkP" VALUES (15, 5, 'Foundations', 750, 'm3', 295.000);
INSERT INTO "WorkP" VALUES (16, 5, 'RC Structure', 3200, 'm2', 210.000);
INSERT INTO "WorkP" VALUES (17, 5, 'Medical Fluids', 1, 'lumpsum', 145000.000);
INSERT INTO "WorkP" VALUES (18, 5, 'Plasterboard', 4000, 'm2', 40.000);
-- SD 6 - Sfax Warehouse
INSERT INTO "WorkP" VALUES (19, 6, 'Concrete Slab', 2000, 'm2', 95.000);
INSERT INTO "WorkP" VALUES (20, 6, 'Steel Frame', 45, 'tonne', 2100.000);
INSERT INTO "WorkP" VALUES (21, 6, 'Sheet Roofing', 2000, 'm2', 55.000);
-- SD 7 - Tunis Lake Tower
INSERT INTO "WorkP" VALUES (22, 7, 'Bored Piles', 200, 'unit', 1800.000);
INSERT INTO "WorkP" VALUES (23, 7, 'Raft Foundation', 1800, 'm2', 280.000);
INSERT INTO "WorkP" VALUES (24, 7, 'RC Shear Walls', 4500, 'm2', 195.000);
```


Populating the table 'Document'

```
INSERT INTO "Document" VALUES (1, 1, 'Contract', '/docs/p1/contract.pdf', TO_DATE('2023-01-25', 'YYYY-MM-DD'), 'Confidential');
INSERT INTO "Document" VALUES (2, 1, 'Plan', '/docs/p1/site_plan.pdf', TO_DATE('2023-02-05', 'YYYY-MM-DD'), 'Restricted');
INSERT INTO "Document" VALUES (3, 1, 'Quote', '/docs/p1/quote.pdf', TO_DATE('2023-02-10', 'YYYY-MM-DD'), 'Restricted');
INSERT INTO "Document" VALUES (4, 1, 'QC Sheet', '/docs/p1/qc_sheet.pdf', TO_DATE('2023-08-01', 'YYYY-MM-DD'), 'Public');
INSERT INTO "Document" VALUES (5, 1, 'Approval', '/docs/p1/approval.pdf', TO_DATE('2023-09-01', 'YYYY-MM-DD'), 'Confidential');
INSERT INTO "Document" VALUES (6, 2, 'Contract', '/docs/p2/contract.pdf', TO_DATE('2023-06-15', 'YYYY-MM-DD'), 'Confidential');
INSERT INTO "Document" VALUES (7, 2, 'Plan', '/docs/p2/floor_plan.pdf', TO_DATE('2023-07-03', 'YYYY-MM-DD'), 'Restricted');
INSERT INTO "Document" VALUES (8, 2, 'Quote', '/docs/p2/quote.pdf', TO_DATE('2023-07-10', 'YYYY-MM-DD'), 'Restricted');
INSERT INTO "Document" VALUES (9, 2, 'QC Sheet', '/docs/p2/qc_sheet.pdf', TO_DATE('2024-01-10', 'YYYY-MM-DD'), 'Public');
INSERT INTO "Document" VALUES (10, 3, 'Contract', '/docs/p3/contract.pdf', TO_DATE('2024-03-20', 'YYYY-MM-DD'), 'Confidential');
INSERT INTO "Document" VALUES (11, 3, 'Plan', '/docs/p3/site_plan.pdf', TO_DATE('2024-04-02', 'YYYY-MM-DD'), 'Restricted');
INSERT INTO "Document" VALUES (12, 4, 'Contract', '/docs/p4/contract.pdf', TO_DATE('2024-09-25', 'YYYY-MM-DD'), 'Confidential');
INSERT INTO "Document" VALUES (13, 4, 'Approval', '/docs/p4/approval.pdf', TO_DATE('2024-10-05', 'YYYY-MM-DD'), 'Restricted');
INSERT INTO "Document" VALUES (14, 4, 'Plan', '/docs/p4/layout_plan.pdf', TO_DATE('2024-10-10', 'YYYY-MM-DD'), 'Restricted');
INSERT INTO "Document" VALUES (15, 5, 'Contract', '/docs/p5/contract.pdf', TO_DATE('2025-02-20', 'YYYY-MM-DD'), 'Confidential');
INSERT INTO "Document" VALUES (16, 5, 'Plan', '/docs/p5/clinic_plan.pdf', TO_DATE('2025-03-05', 'YYYY-MM-DD'), 'Restricted');
INSERT INTO "Document" VALUES (17, 6, 'Contract', '/docs/p6/contract.pdf', TO_DATE('2025-08-10', 'YYYY-MM-DD'), 'Confidential');
INSERT INTO "Document" VALUES (18, 7, 'Contract', '/docs/p7/contract.pdf', TO_DATE('2026-01-15', 'YYYY-MM-DD'), 'Confidential');
INSERT INTO "Document" VALUES (19, 7, 'Plan', '/docs/p7/tower_plan.pdf', TO_DATE('2026-02-03', 'YYYY-MM-DD'), 'Restricted');
INSERT INTO "Document" VALUES (20, 7, 'Quote', '/docs/p7/quote.pdf', TO_DATE('2026-02-08', 'YYYY-MM-DD'), 'Restricted');
INSERT INTO "Document" VALUES (21, 8, 'Contract', '/docs/p8/contract.pdf', TO_DATE('2026-03-01', 'YYYY-MM-DD'), 'Confidential');
INSERT INTO "Document" VALUES (22, 8, 'Plan', '/docs/p8/school_plan.pdf', TO_DATE('2026-03-18', 'YYYY-MM-DD'), 'Public');
INSERT INTO "Document" VALUES (23, 9, 'Contract', '/docs/p9/contract.pdf', TO_DATE('2026-01-10', 'YYYY-MM-DD'), 'Confidential');
INSERT INTO "Document" VALUES (24, 10, 'Contract', '/docs/p10/contract.pdf', TO_DATE('2025-04-20', 'YYYY-MM-DD'), 'Confidential');
INSERT INTO "Document" VALUES (25, 10, 'QC Sheet', '/docs/p10/qc_sheet.pdf', TO_DATE('2025-09-10', 'YYYY-MM-DD'), 'Public');
INSERT INTO "Document" VALUES (26, 11, 'Contract', '/docs/p11/contract.pdf', TO_DATE('2023-12-20', 'YYYY-MM-DD'), 'Confidential');
INSERT INTO "Document" VALUES (27, 12, 'Contract', '/docs/p12/contract.pdf', TO_DATE('2023-05-15', 'YYYY-MM-DD'), 'Confidential');
INSERT INTO "Document" VALUES (28, 12, 'Plan', '/docs/p12/pool_plan.pdf', TO_DATE('2023-06-01', 'YYYY-MM-DD'), 'Restricted');
INSERT INTO "Document" VALUES (29, 13, 'Contract', '/docs/p13/contract.pdf', TO_DATE('2024-05-20', 'YYYY-MM-DD'), 'Confidential');
INSERT INTO "Document" VALUES (30, 14, 'Contract', '/docs/p14/contract.pdf', TO_DATE('2024-12-15', 'YYYY-MM-DD'), 'Confidential');
INSERT INTO "Document" VALUES (31, 15, 'Contract', '/docs/p15/contract.pdf', TO_DATE('2026-01-25', 'YYYY-MM-DD'), 'Confidential');
```

Populating the table 'Situation'

```
-- Project 1 - Carthage Residency
INSERT INTO "Situation" VALUES (1, 1, 'Earthworks and foundations', TO_DATE('2023-02-01', 'YYYY-MM-DD'), TO_DATE('2023-02-28', 'YYYY-MM-DD'));
INSERT INTO "Situation" VALUES (2, 1, 'Reinforced concrete 60%', TO_DATE('2023-03-01', 'YYYY-MM-DD'), TO_DATE('2023-03-31', 'YYYY-MM-DD'));
INSERT INTO "Situation" VALUES (3, 1, 'Masonry floors R+1 and R+2', TO_DATE('2023-06-01', 'YYYY-MM-DD'), TO_DATE('2023-06-30', 'YYYY-MM-DD'));
INSERT INTO "Situation" VALUES (4, 1, 'Coating and finishing works', TO_DATE('2023-10-01', 'YYYY-MM-DD'), TO_DATE('2023-10-31', 'YYYY-MM-DD'));
-- Project 2 - Sousse Beach Hotel
INSERT INTO "Situation" VALUES (5, 2, 'Foundations completed', TO_DATE('2023-07-01', 'YYYY-MM-DD'), TO_DATE('2023-07-31', 'YYYY-MM-DD'));
INSERT INTO "Situation" VALUES (6, 2, 'RC structure floors 1 and 2', TO_DATE('2023-09-01', 'YYYY-MM-DD'), TO_DATE('2023-09-30', 'YYYY-MM-DD'));
INSERT INTO "Situation" VALUES (7, 2, 'Joinery and painting start', TO_DATE('2024-01-01', 'YYYY-MM-DD'), TO_DATE('2024-01-31', 'YYYY-MM-DD'));
INSERT INTO "Situation" VALUES (8, 2, 'Interior painting completed', TO_DATE('2024-04-01', 'YYYY-MM-DD'), TO_DATE('2024-04-30', 'YYYY-MM-DD'));
-- Project 3 - Bizerte Complex
INSERT INTO "Situation" VALUES (9, 3, 'General earthworks', TO_DATE('2024-04-01', 'YYYY-MM-DD'), TO_DATE('2024-04-30', 'YYYY-MM-DD'));
INSERT INTO "Situation" VALUES (10, 3, 'Concrete slab 50% progress', TO_DATE('2024-07-01', 'YYYY-MM-DD'), TO_DATE('2024-07-31', 'YYYY-MM-DD'));
INSERT INTO "Situation" VALUES (11, 3, 'Roof covering completed', TO_DATE('2025-01-01', 'YYYY-MM-DD'), TO_DATE('2025-01-31', 'YYYY-MM-DD'));
-- Project 4 - Ariana Tech Park
INSERT INTO "Situation" VALUES (12, 4, 'Earthworks and concrete start', TO_DATE('2024-10-01', 'YYYY-MM-DD'), TO_DATE('2024-10-31', 'YYYY-MM-DD'));
INSERT INTO "Situation" VALUES (13, 4, 'RC structure and networks', TO_DATE('2025-02-01', 'YYYY-MM-DD'), TO_DATE('2025-02-28', 'YYYY-MM-DD'));
INSERT INTO "Situation" VALUES (14, 4, 'Finishing and handover', TO_DATE('2025-07-01', 'YYYY-MM-DD'), TO_DATE('2025-07-31', 'YYYY-MM-DD'));
-- Project 5 - Hammamet Clinic
INSERT INTO "Situation" VALUES (15, 5, 'Foundation phase 1', TO_DATE('2025-03-01', 'YYYY-MM-DD'), TO_DATE('2025-03-31', 'YYYY-MM-DD'));
INSERT INTO "Situation" VALUES (16, 5, 'RC structure ground floor', TO_DATE('2025-06-01', 'YYYY-MM-DD'), TO_DATE('2025-06-30', 'YYYY-MM-DD'));
INSERT INTO "Situation" VALUES (17, 5, 'Partitions and fluids start', TO_DATE('2025-10-01', 'YYYY-MM-DD'), TO_DATE('2025-10-31', 'YYYY-MM-DD'));
-- Project 6 - Sfax Warehouse
INSERT INTO "Situation" VALUES (18, 6, 'Concrete slab 1000 sqm poured', TO_DATE('2025-09-01', 'YYYY-MM-DD'), TO_DATE('2025-09-30', 'YYYY-MM-DD'));
INSERT INTO "Situation" VALUES (19, 6, 'Steel frame and roof covering', TO_DATE('2025-12-01', 'YYYY-MM-DD'), TO_DATE('2025-12-31', 'YYYY-MM-DD'));
-- Project 7 - Tunis Lake Tower
INSERT INTO "Situation" VALUES (20, 7, 'Bored piles 80 units done', TO_DATE('2026-02-01', 'YYYY-MM-DD'), TO_DATE('2026-02-28', 'YYYY-MM-DD'));
INSERT INTO "Situation" VALUES (21, 7, 'Raft foundation started', TO_DATE('2026-03-01', 'YYYY-MM-DD'), TO_DATE('2026-03-31', 'YYYY-MM-DD'));
```


Populating the table 'Achieve'

```
-- Project 1 (SD1: WorkP 1-4)
INSERT INTO "Achieve" VALUES (1, 1, 280);
INSERT INTO "Achieve" VALUES (1, 2, 150);
INSERT INTO "Achieve" VALUES (2, 1, 500);
INSERT INTO "Achieve" VALUES (2, 2, 550);
INSERT INTO "Achieve" VALUES (2, 3, 600);
INSERT INTO "Achieve" VALUES (3, 2, 800);
INSERT INTO "Achieve" VALUES (3, 3, 2400);
INSERT INTO "Achieve" VALUES (3, 4, 900);
INSERT INTO "Achieve" VALUES (4, 3, 2400);
INSERT INTO "Achieve" VALUES (4, 4, 1800);
-- Project 2 (SD2: WorkP 5-8)
INSERT INTO "Achieve" VALUES (5, 5, 400);
INSERT INTO "Achieve" VALUES (5, 6, 200);
INSERT INTO "Achieve" VALUES (6, 5, 700);
INSERT INTO "Achieve" VALUES (6, 6, 800);
INSERT INTO "Achieve" VALUES (7, 6, 1100);
INSERT INTO "Achieve" VALUES (7, 7, 300);
INSERT INTO "Achieve" VALUES (7, 8, 1000);
INSERT INTO "Achieve" VALUES (8, 7, 480);
INSERT INTO "Achieve" VALUES (8, 8, 4500);
-- Project 3 (SD3: WorkP 9-11)
INSERT INTO "Achieve" VALUES (9, 9, 250);
INSERT INTO "Achieve" VALUES (9, 10, 100);
INSERT INTO "Achieve" VALUES (10, 9, 400);
INSERT INTO "Achieve" VALUES (10, 10, 380);
INSERT INTO "Achieve" VALUES (11, 10, 560);
INSERT INTO "Achieve" VALUES (11, 11, 900);
```

Conclusion

Once the physical data model has been established and the database created, our attention in the next chapter will focus on querying the database, formulating SQL queries, and developing web interfaces.

Chapter 4. Presentation and Execution of SQL Queries

Introduction

Following the technical creation of the database in the previous chapter, this part of our study focuses on transforming the functional requirements into operational data operations. This chapter plays an essential role in demonstrating that the designed database model is capable of satisfying the core management objectives defined at the beginning of the project. To achieve this, we first introduce the various SQL queries developed to meet the specific needs of construction site tracking. These queries are then formally presented using SQL language, and the obtained results are analyzed to verify their accuracy and coherence with the underlying data.

1. Presentation of Queries

Here is the complete organized list of all 23 queries:

1. Display all documents with a 'Confidential' authorization level.
2. List all projects with their client's full name and specification document version.
3. Display all work packages that have never been achieved in any situation.
4. Display the number of work packages per specification document, ordered by count descending.
5. List clients who have ordered more than one project.
6. For each project, display the total calculated cost of its work packages along with the project budget, ordered by total cost descending.
7. Display all projects with their remaining days and a priority level: HIGH if less than 30 days, MEDIUM if between 30 and 90 days, LOW otherwise.
8. Display all work packages whose unit price is above the average unit price across all work packages.
9. List all situations that include at least one of these work packages: 'Earthworks', 'Foundations', or 'RC Structure'.

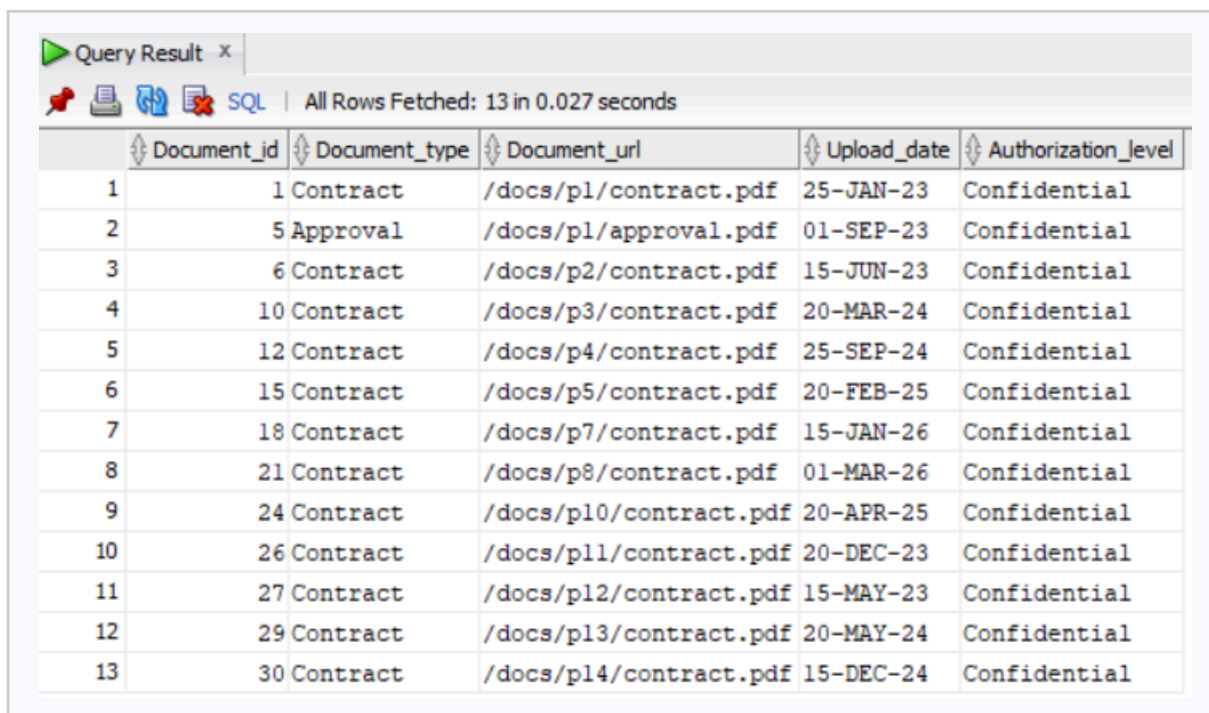
10. Display all clients who have not ordered any project with a budget exceeding 1,000,000.
11. List all projects that have at least one recorded situation.
12. Display a combined list of all active project names and all client full names as a single unified list.
13. Find clients who have both a project with a budget above 500,000 AND a project with a budget below 500,000.
14. Display specification documents that contain work packages but are not linked to any project.
15. For each work package, display the total achieved quantity compared to the expected quantity and the remaining quantity.
16. For each specification document, rank its work packages by unit price from highest to lowest.
17. Find work packages with inconsistent progress reporting by showing both the sample standard deviation and population standard deviation of achieved quantities, filtered to those with a standard deviation above 100.
18. Display all projects along with their total number of situations and total number of documents.
19. Display clients whose total project budget exceeds the overall average total budget per client.
20. For each specification document, display only the work package with the highest unit price.
21. Display projects where the achieved quantity in any single situation exceeds the expected quantity of the corresponding work package, showing the excess amount.
22. Display a full progress report per project and work package, showing expected quantity, total achieved, remaining quantity, completion percentage, and earned value.

23. Generate a complete financial dashboard per project: total calculated cost, total earned value from achievements, budget consumption percentage, remaining budget, and a financial health status evaluated as OVER BUDGET, AT RISK, or ON TRACK.

2. SQL formulation and execution

Query 1: Display all documents with a 'Confidential' authorization level.

```
SELECT
    d."Document_id",
    d."Document_type",
    d."Document_url",
    d."Upload_date",
    d."Authorization_level"
FROM "Document" d
WHERE d."Authorization_level" = 'Confidential';
```



The screenshot shows a database query result interface. At the top, there's a green play button icon and the text 'Query Result x'. Below it, there are icons for a red pin, a document, a blue refresh icon, and a red SQL icon, followed by the text 'All Rows Fetched: 13 in 0.027 seconds'. The main part of the image is a table with 6 columns: 'Document_id', 'Document_type', 'Document_url', 'Upload_date', and 'Authorization_level'. The table contains 13 rows of data, all with 'Confidential' in the 'Authorization_level' column. The 'Document_type' column shows a mix of 'Contract' and 'Approval'.

	Document_id	Document_type	Document_url	Upload_date	Authorization_level
1	1	Contract	/docs/p1/contract.pdf	25-JAN-23	Confidential
2	5	Approval	/docs/p1/approval.pdf	01-SEP-23	Confidential
3	6	Contract	/docs/p2/contract.pdf	15-JUN-23	Confidential
4	10	Contract	/docs/p3/contract.pdf	20-MAR-24	Confidential
5	12	Contract	/docs/p4/contract.pdf	25-SEP-24	Confidential
6	15	Contract	/docs/p5/contract.pdf	20-FEB-25	Confidential
7	18	Contract	/docs/p7/contract.pdf	15-JAN-26	Confidential
8	21	Contract	/docs/p8/contract.pdf	01-MAR-26	Confidential
9	24	Contract	/docs/p10/contract.pdf	20-APR-25	Confidential
10	26	Contract	/docs/p11/contract.pdf	20-DEC-23	Confidential
11	27	Contract	/docs/p12/contract.pdf	15-MAY-23	Confidential
12	29	Contract	/docs/p13/contract.pdf	20-MAY-24	Confidential
13	30	Contract	/docs/p14/contract.pdf	15-DEC-24	Confidential

Query 2: List all projects with their client's full name and specification document version.

```
SELECT
    p."Project_name",
    c."First_name" || ' ' || c."Last_name" AS Client_Full_Name,
    sd."Version" AS SD_Version
FROM "Project" p
JOIN "Client" c ON p."Client_id" = c."Client_id"
JOIN SD sd ON p."SD_id" = sd."SD_id";
```

Query Result x

All Rows Fetched: 12 in 0.008 seconds

	Project_name	CLIENT_FULL_NAME	SD_VERSION
1	Carthage Residency	Ahmed Ben Salah	v1.0
2	Sousse Beach Hotel	Fatma Trabelsi	v2.1
3	Bizerte Complex	Karim Gharbi	v1.0
4	Ariana Tech Park	Leila Chaabane	v1.2
5	Hammamet Clinic	Amine Missaoui	v1.0
6	Tunis Lake Tower	Riadh Jebali	v1.0
7	Kairouan School	Nadia Bouaziz	v1.1
8	Nabeul North Villa	Slim Zouari	v1.0
9	Carthage Annex B	Ahmed Ben Salah	v1.1
10	Sousse Swimming Pool	Fatma Trabelsi	v1.0
11	Industrial Zone Exp.	Karim Gharbi	v1.0
12	Hammamet Apartment	Amine Missaoui	v1.2

Query 3: Display all work packages that have never been achieved in any situation.

```
SELECT
    wp."WorkP_name",
    wp."Expected_quantity",
    wp."Measurment_unit",
    wp."Unit_price"
FROM "WorkP" wp
LEFT JOIN "Achieve" a ON wp."WorkP_id" = a."WorkP_id"
WHERE a."WorkP_id" IS NULL;
```

Query Result x			
SQL All Rows Fetched: 11 in 0.006 seconds			
	WorkP_name	Expected_quantity	Measurment_unit
1	Green Spaces	600 m2	25
2	RC Shear Walls	1200 m2	185
3	Sheet Roofing	2000 m2	55
4	Interior Paint	1200 m2	15
5	Finishing Works	1 lumpsum	55000
6	Finishing Works	1 lumpsum	85000
7	Bored Piles	120 unit	1800
8	Steel Frame	45 tonne	2100
9	Parking Raft	800 m2	260
10	Concrete Slab	2000 m2	95
11	Shell Works	1 lumpsum	90000

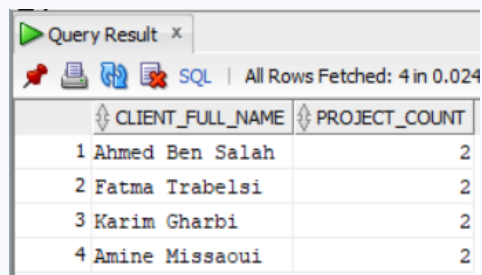
Query 4: Display the number of work packages per specification document, ordered by count descending.

```
SELECT
    sd."SD_id",
    sd."Version",
    COUNT(wp."WorkP_id") AS WorkPackage_Count
FROM SD sd
LEFT JOIN "WorkP" wp ON sd."SD_id" = wp."SD_id"
GROUP BY
    sd."SD_id",
    sd."Version"
ORDER BY WorkPackage_Count DESC;
```

Query Result x			
SQL All Rows Fetched: 15 in 0.027 s			
	SD_id	Version	WORKPACKAGE_COUNT
1	2	v2.1	4
2	13	v1.0	4
3	1	v1.0	4
4	5	v1.0	4
5	15	v1.0	3
6	9	v1.0	3
7	10	v1.0	3
8	11	v1.1	3
9	12	v1.0	3
10	14	v1.2	3
11	7	v1.0	3
12	6	v2.0	3
13	4	v1.2	3
14	3	v1.0	3
15	8	v1.1	3

Query 5: List clients who have ordered more than one project.

```
SELECT
    c."First_name" || ' ' || c."Last_name" AS Client_Full_Name,
    COUNT(p."Project_id") AS Project_Count
FROM "Client" c
JOIN "Project" p ON c."Client_id" = p."Client_id"
GROUP BY
    c."Client_id",
    c."First_name",
    c."Last_name"
HAVING COUNT(p."Project_id") > 1;
```



The screenshot shows a 'Query Result' window with the following data:

	CLIENT_FULL_NAME	PROJECT_COUNT
1	Ahmed Ben Salah	2
2	Fatma Trabelsi	2
3	Karim Gharbi	2
4	Amine Missaoui	2

Query 6: For each project, display the total calculated cost of its work packages along with the project budget, ordered by total cost descending.

```
SELECT
    p."Project_name",
    p."Budget",
    SUM(wp."Unit_price" * wp."Expected_quantity") AS Total_Calculated_Cost
FROM "Project" p
JOIN SD sd ON p."SD_id" = sd."SD_id"
JOIN "WorkP" wp ON wp."SD_id" = sd."SD_id"
GROUP BY
    p."Project_id",
    p."Project_name",
    p."Budget"
ORDER BY Total_Calculated_Cost DESC;
```

Query Result x

 | All Rows Fetched: 12 in 0.021 seconds

	Project_name	Budget	TOTAL_CALCULATED_COST
1	Tunis Lake Tower	1950000	1741500
2	Hammamet Clinic	1100000	1198250
3	Sousse Beach Hotel	1850000	731300
4	Industrial Zone Exp.	850000	703200
5	Carthage Residency	1250000	550900
6	Ariana Tech Park	780000	503400
7	Hammamet Apartment	630000	503000
8	Bizerte Complex	975000	271100
9	Nabeul North Villa	480000	233400
10	Kairouan School	310000	220000
11	Carthage Annex B	390000	189300
12	Sousse Swimming Pool	220000	118200

Query 7: Display all projects with their remaining days and a priority level: HIGH if less than 30 days, MEDIUM if between 30 and 90 days, LOW otherwise.

```

SELECT
  p."Project_name",
  ROUND(p."Project_expected_end_date" - SYSDATE) AS Remaining_Days,
  CASE
    WHEN p."Project_expected_end_date" - SYSDATE < 30 THEN 'HIGH'
    WHEN p."Project_expected_end_date" - SYSDATE < 90 THEN 'MEDIUM'
    ELSE 'LOW'
  END AS Priority_Level
FROM "Project" p;

```


Query Result x		
SQL All Rows Fetched: 12 in 0.007 seconds		
	Project_name	REMAINING_DAYS PRIORITY_LEVEL
1	Carthage Residency	-655 HIGH
2	Sousse Beach Hotel	-290 HIGH
3	Bizerte Complex	-16 HIGH
4	Ariana Tech Park	-198 HIGH
5	Hammamet Clinic	318 LOW
6	Tunis Lake Tower	655 LOW
7	Kairouan School	440 LOW
8	Nabeul North Villa	137 LOW
9	Carthage Annex B	-471 HIGH
10	Sousse Swimming Pool	-685 HIGH
11	Industrial Zone Exp.	45 MEDIUM
12	Hammamet Apartment	-106 HIGH

Query 8: Display all work packages whose unit price is above the average unit price across all work packages.

```

SELECT
    wp."WorkP_name",
    wp."Measurment_unit",
    wp."Unit_price"
FROM "WorkP" wp
WHERE wp."Unit_price" > (
    SELECT AVG(wp2."Unit_price")
    FROM "WorkP" wp2
);

```

Query Result x

SQL | All Rows Fetched: 10 in 0.008 seconds

	WorkP_name	Measurment_unit	Unit_price
1	Cable Networks	lumpsum	72000
2	Medical Fluids	lumpsum	145000
3	Shell Works	lumpsum	120000
4	Finishing Works	lumpsum	85000
5	Shell Works	lumpsum	180000
6	Utility Networks	lumpsum	95000
7	Infrastructure	lumpsum	160000
8	Superstructure	lumpsum	280000
9	Shell Works	lumpsum	90000
10	Finishing Works	lumpsum	55000

Query 9: List all situations that include at least one of these work packages: 'Earthworks', 'Foundations', or 'RC Structure'.

```
SELECT DISTINCT
    s."Situation_id",
    s."Comments",
    s."Start_date",
    s."End_date"
FROM "Situation" s
JOIN "Achieve" a ON s."Situation_id" = a."Situation_id"
JOIN "WorkP" wp ON a."WorkP_id" = wp."WorkP_id"
WHERE wp."WorkP_name" IN ('Earthworks', 'Foundations', 'RC Structure');
```

Query Result x				
SQL All Rows Fetched: 17 in 0.014 seconds				
	Situation_id	Comments	Start_date	End_date
1	1	Earthworks and foundations	01-FEB-23	28-FEB-23
2	2	Reinforced concrete 60%	01-MAR-23	31-MAR-23
3	5	Foundations completed	01-JUL-23	31-JUL-23
4	6	RC structure floors 1 and 2	01-SEP-23	30-SEP-23
5	7	Joinery and painting start	01-JAN-24	31-JAN-24
6	9	General earthworks	01-APR-24	30-APR-24
7	10	Concrete slab 50% progress	01-JUL-24	31-JUL-24
8	12	Earthworks and concrete start	01-OCT-24	31-OCT-24
9	13	RC structure and networks	01-FEB-25	28-FEB-25
10	14	Finishing and handover	01-JUL-25	31-JUL-25
11	15	Foundation phase 1	01-MAR-25	31-MAR-25
12	16	RC structure ground floor	01-JUN-25	30-JUN-25
13	17	Partitions and fluids start	01-OCT-25	31-OCT-25
14	27	Earthworks and foundations	01-JAN-24	31-JAN-24
15	28	Masonry and finishing works	01-JUN-24	30-JUN-24
16	32	Earthworks phase 1	01-JUN-24	30-JUN-24
17	33	Industrial slab and structure	01-OCT-24	31-OCT-24

Query 10: Display all clients who have not ordered any project with a budget exceeding 1,000,000.

```

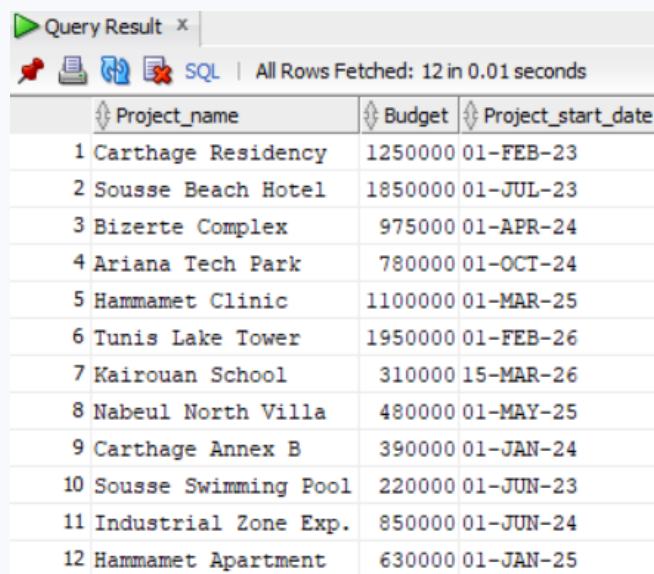
SELECT
    c."First_name" || ' ' || c."Last_name" AS Client_Full_Name,
    c."Address"
FROM "Client" c
WHERE c."Client_id" NOT IN (
    SELECT p."Client_id"
    FROM "Project" p
    WHERE p."Budget" > 1000000
);

```

Query Result x		
SQL All Rows Fetched: 4 in 0.007 seconds		
	CLIENT_FULL_NAME	Address
1	Nadia Bouaziz	9 Tunis Avenue, Kairouan
2	Leila Chaabane	22 Republic Street, Nabeul
3	Slim Zouari	77 Bechir Sfar Avenue, Sfax
4	Karim Gharbi	8 Ibn Khaldoun Street, Sousse

Query 11: List all projects that have at least one recorded situation.

```
SELECT
    p."Project_name",
    p."Budget",
    p."Project_start_date"
FROM "Project" p
WHERE EXISTS (
    SELECT 1
    FROM "Situation" s
    WHERE s."Project_id" = p."Project_id"
);
```



The screenshot shows a 'Query Result' window with 12 rows of data. The columns are 'Project_name', 'Budget', and 'Project_start_date'. The data is as follows:

	Project_name	Budget	Project_start_date
1	Carthage Residency	1250000	01-FEB-23
2	Sousse Beach Hotel	1850000	01-JUL-23
3	Bizerte Complex	975000	01-APR-24
4	Ariana Tech Park	780000	01-OCT-24
5	Hammamet Clinic	1100000	01-MAR-25
6	Tunis Lake Tower	1950000	01-FEB-26
7	Kairouan School	310000	15-MAR-26
8	Nabeul North Villa	480000	01-MAY-25
9	Carthage Annex B	390000	01-JAN-24
10	Sousse Swimming Pool	220000	01-JUN-23
11	Industrial Zone Exp.	850000	01-JUN-24
12	Hammamet Apartment	630000	01-JAN-25

Query 12: Display a combined list of all project names labeled as 'Active' if their end date is in the future, and all client full names labeled as 'Client', as a single unified list.

```
SELECT
    p."Project_name" AS Name,
    'Active Project' AS Type
FROM "Project" p
WHERE p."Project_expected_end_date" > SYSDATE
UNION
SELECT
    c."First_name" || ' ' || c."Last_name",
    'Client'
FROM "Client" c;
```

Query Result x		
All Rows Fetched: 13 in 0.011 sec		
	NAME	TYPE
1	Hammamet Clinic	Active Project
2	Tunis Lake Tower	Active Project
3	Kairouan School	Active Project
4	Nabeul North Villa	Active Project
5	Industrial Zone Exp.	Active Project
6	Ahmed Ben Salah	Client
7	Fatma Trabelsi	Client
8	Karim Gharbi	Client
9	Leila Chaabane	Client
10	Amine Missaoui	Client
11	Riadh Jebali	Client
12	Nadia Bouaziz	Client
13	Slim Zouari	Client

Query 13: Find clients who have both a project with a budget above 500,000 AND a project with a budget below 500,000.

```
SELECT c."Client_id", c."First_name", c."Last_name"
FROM "Client" c
JOIN "Project" p ON c."Client_id" = p."Client_id"
WHERE p."Budget" > 500000
INTERSECT
SELECT c."Client_id", c."First_name", c."Last_name"
FROM "Client" c
JOIN "Project" p ON c."Client_id" = p."Client_id"
WHERE p."Budget" < 500000;
```

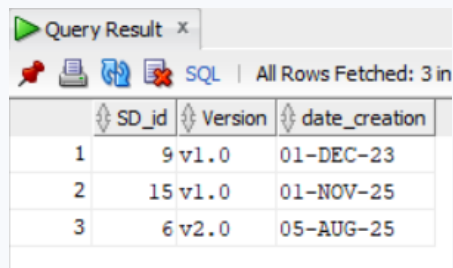
Query Result x

SQL | All Rows Fetched: 2 in 0.0

	Client_id	First_name	Last_name
1	1	Ahmed	Ben Salah
2	2	Fatma	Trabelsi

Query 14: Display specification documents that contain work packages but are not linked to any project.

```
SELECT sd."SD_id", sd."Version", sd."date_creation"
FROM SD sd
JOIN "WorkP" wp ON sd."SD_id" = wp."SD_id"
MINUS
SELECT sd."SD_id", sd."Version", sd."date_creation"
FROM SD sd
JOIN "Project" p ON sd."SD_id" = p."SD_id";
```



Query Result x

SQL | All Rows Fetched: 3 in

	SD_id	Version	date_creation
1	9	v1.0	01-DEC-23
2	15	v1.0	01-NOV-25
3	6	v2.0	05-AUG-25

Query 15: For each work package, display the total achieved quantity compared to the expected quantity and the remaining quantity.

```
SELECT
  wp."WorkP_name",
  wp."Expected_quantity",
  (SELECT NVL(SUM(a."Achieved_quantity"), 0)
   FROM "Achieve" a
   WHERE a."WorkP_id" = wp."WorkP_id") AS Total_Achieved,
  wp."Expected_quantity" -
  (SELECT NVL(SUM(a."Achieved_quantity"), 0)
   FROM "Achieve" a
   WHERE a."WorkP_id" = wp."WorkP_id") AS Remaining_Quantity
FROM "WorkP" wp;
```

WorkP_name	Expected_quantity	TOTAL_ACHIEVED	REMAINING_QUANTITY
1 Earthworks	500	780	-280
2 Reinforced Con.	800	1500	-700
3 Masonry	2400	5400	-3000
4 Facade Coating	1800	2700	-900
5 Foundations	700	1100	-400
6 RC Structure	1100	2100	-1000
7 Alu. Joinery	480	780	-300
8 Interior Paint	4500	5500	-1000
9 Earthworks	400	650	-250
10 Slab Concrete	560	1040	-480
11 Roof Cover	900	900	0
12 Earthworks	900	1500	-600
13 Reinforced Con.	1200	1030	170
14 Cable Networks	1	1	0
15 Foundations	750	1150	-400
16 RC Structure	3200	6200	-3000
17 Medical Fluids	1	1	0
18 Plasterboard	4000	1000	3000
19 Bored Piles	200	280	-80
20 Raft Foundation	1800	1200	600
21 RC Shear Walls	4500	500	4000
22 Shell Works	1	1	0
23 Shell Works	1	2	-1
24 Floor Tiling	450	650	-200

Query 16: For each specification document, rank its work packages by unit price from highest to lowest.

```

SELECT
    sd."Version"      AS SD_Version,
    wp."WorkP_name",
    wp."Unit_price",
    wp."Measurment_unit",
    RANK() OVER (
        PARTITION BY sd."SD_id"
        ORDER BY wp."Unit_price" DESC
    ) AS Price_Rank
FROM "WorkP" wp
JOIN SD sd ON wp."SD_id" = sd."SD_id";

```

Query Result x				
All Rows Fetched: 49 in 0.008 seconds				
SD_VERSION	WorkP_name	Unit_price	Measurment_unit	PRICE_RANK
1 v1.0	Reinforced Con.	320 m3		1
2 v1.0	Masonry	85 m2		2
3 v1.0	Earthworks	45 m3		3
4 v1.0	Facade Coating	38 m2		4
5 v2.1	RC Structure	315 m3		1
6 v2.1	Foundations	290 m3		2
7 v2.1	Alu. Joinery	210 m1		3
8 v2.1	Interior Paint	18 m2		4
9 v1.0	Slab Concrete	300 m3		1
10 v1.0	Roof Cover	95 m2		2
11 v1.0	Earthworks	44 m3		3
12 v1.2	Cable Networks	72000 lumpsum		1
13 v1.2	Reinforced Con.	325 m3		2
14 v1.2	Earthworks	46 m3		3
15 v1.0	Medical Fluids	145000 lumpsum		1
16 v1.0	Foundations	295 m3		2
17 v1.0	RC Structure	210 m2		3
18 v1.0	Plasterboard	40 m2		4
19 v2.0	Steel Frame	2100 tonne		1
20 v2.0	Concrete Slab	95 m2		2
21 v2.0	Sheet Roofing	55 m2		3
22 v1.0	Bored Piles	1800 unit		1
23 v1.0	Raft Foundation	280 m2		2
24 v1.0	RC Shear Walls	195 m2		3
25 v1.1	Shell Works	120000 lumpsum		1
26 v1.1	Finishing Works	85000 lumpsum		2

Query 17: Find work packages with inconsistent progress reporting by showing both the sample standard deviation and population standard deviation of achieved quantities, filtered to those with a standard deviation above 100.

```

SELECT
    wp."WorkP_name",
    wp."Measurment_unit",
    COUNT(a."Situation_id")           AS Situation_Count,
    ROUND(AVG(a."Achieved_quantity"), 2) AS Avg_Achieved,
    ROUND(STDDEV(a."Achieved_quantity"), 2) AS StdDev_Sample,
    ROUND(STDDEV_POP(a."Achieved_quantity"), 2) AS StdDev_Population
FROM "WorkP" wp
JOIN "Achieve" a ON wp."WorkP_id" = a."WorkP_id"
GROUP BY
    wp."WorkP_id",
    wp."WorkP_name",
    wp."Measurment_unit"
HAVING
    COUNT(a."Situation_id") > 1
    AND STDDEV(a."Achieved_quantity") > 100;

```


Query Result x						
All Rows Fetched: 19 in 0.016 seconds						
	WorkP_name	Measurment_unit	SITUATION_COUNT	AVG_ACHIEVED	STDDEV_SAMPLE	STDDEV_POPULATION
1	Earthworks	m3	2	390	155.56	110
2	Reinforced Con.	m3	3	500	327.87	267.71
3	Masonry	m2	3	1800	1039.23	848.53
4	Facade Coating	m2	2	1350	636.4	450
5	Foundations	m3	2	550	212.13	150
6	RC Structure	m3	3	700	458.26	374.17
7	Alu. Joinery	m1	2	390	127.28	90
8	Interior Paint	m2	2	2750	2474.87	1750
9	Earthworks	m3	2	325	106.07	75
10	Slab Concrete	m3	3	346.67	231.8	189.27
11	Earthworks	m3	3	500	400	326.6
12	Reinforced Con.	m3	3	343.33	482.32	393.81
13	Foundations	m3	2	575	247.49	175
14	RC Structure	m2	3	2066.67	1205.54	984.32
15	Raft Foundation	m2	2	600	424.26	300
16	Floor Tiling	m2	2	325	176.78	125
17	Reinforced Con.	m3	2	265	120.21	85
18	Earthworks	m3	2	900	424.26	300
19	Industrial Slab	m2	3	2266.67	1365.04	1114.55

Query 18: Display all projects along with their total number of situations and total number of documents.

```

WITH SituationCount AS (
    SELECT "Project_id", COUNT(*) AS Total_Situations
    FROM "Situation"
    GROUP BY "Project_id"
),
DocumentCount AS (
    SELECT "Project_id", COUNT(*) AS Total_Documents
    FROM "Document"
    GROUP BY "Project_id"
)
SELECT
    p."Project_name",
    NVL(sc.Total_Situations, 0) AS Total_Situations,
    NVL(dc.Total_Documents, 0) AS Total_Documents
FROM "Project" p
LEFT JOIN SituationCount sc ON p."Project_id" = sc."Project_id"
LEFT JOIN DocumentCount dc ON p."Project_id" = dc."Project_id";

```

Query Result x			
SQL All Rows Fetched: 12 in 0.01 seconds			
Project_name	TOTAL_SITUATIONS	TOTAL_DOCUMENTS	
1 Carthage Residency	4	5	
2 Sousse Beach Hotel	4	4	
3 Bizerte Complex	3	2	
4 Ariana Tech Park	3	3	
5 Hammamet Clinic	3	2	
6 Tunis Lake Tower	2	3	
7 Kairouan School	1	2	
8 Nabeul North Villa	2	2	
9 Carthage Annex B	2	1	
10 Sousse Swimming Pool	3	2	
11 Industrial Zone Exp.	3	1	
12 Hammamet Apartment	2	1	

Query 19: Display clients whose total project budget exceeds the overall average total budget per client.

```

WITH ClientBudget AS (
    SELECT
        c."Client_id",
        c."First_name" || ' ' || c."Last_name" AS Full_Name,
        SUM(p."Budget") AS Total_Budget
    FROM "Client" c
    JOIN "Project" p ON c."Client_id" = p."Client_id"
    GROUP BY c."Client_id", c."First_name", c."Last_name"
),
GlobalAvg AS (
    SELECT AVG(Total_Budget) AS Avg_Budget
    FROM ClientBudget
)
SELECT
    cb.Full_Name,
    cb.Total_Budget,
    ROUND(ga.Avg_Budget, 2) AS Global_Average
FROM ClientBudget cb
CROSS JOIN GlobalAvg ga
WHERE cb.Total_Budget > ga.Avg_Budget;

```

Query Result x			
SQL All Rows Fetched: 5 in 0.021 seconds			
FULL_NAME	TOTAL_BUDGET	GLOBAL_AVERAGE	
1 Ahmed Ben Salah	1640000	1348125	
2 Fatma Trabelsi	2070000	1348125	
3 Karim Gharbi	1825000	1348125	
4 Amine Missaoui	1730000	1348125	
5 Riadh Jebali	1950000	1348125	

Query 20: For each specification document, display only the work package with the highest unit price.

```
WITH RankedWorkP AS (
    SELECT
        wp."WorkP_name",
        wp."Unit_price",
        wp."Measurment_unit",
        sd."Version",
        p."Project_name",
        RANK() OVER (
            PARTITION BY sd."SD_id"
            ORDER BY wp."Unit_price" DESC
        ) AS Price_Rank
    FROM "WorkP" wp
    JOIN SD sd      ON wp."SD_id" = sd."SD_id"
    JOIN "Project" p ON p."SD_id" = sd."SD_id"
)
SELECT
    "Project_name",
    "Version"      AS SD_Version,
    "WorkP_name" AS Most_Expensive_WorkP,
    "Measurment_unit",
    "Unit_price"
FROM RankedWorkP
WHERE Price_Rank = 1;
```

Query Result x

All Rows Fetched: 12 in 0.015 seconds

	Project_name	SD_VERSION	MOST_EXPENSIVE_WORKP	Measurment_unit	Unit_price
1	Carthage Residency	v1.0	Reinforced Con.	m3	320
2	Sousse Beach Hotel	v2.1	RC Structure	m3	315
3	Bizerte Complex	v1.0	Slab Concrete	m3	300
4	Ariana Tech Park	v1.2	Cable Networks	lumpsum	72000
5	Hammamet Clinic	v1.0	Medical Fluids	lumpsum	145000
6	Tunis Lake Tower	v1.0	Bored Piles	unit	1800
7	Kairouan School	v1.1	Shell Works	lumpsum	120000
8	Nabeul North Villa	v1.0	Shell Works	lumpsum	180000
9	Carthage Annex B	v1.1	Reinforced Con.	m3	310
10	Sousse Swimming Pool	v1.0	RC Walls Floor	m2	195
11	Industrial Zone Exp.	v1.0	Utility Networks	lumpsum	95000
12	Hammamet Apartment	v1.2	Superstructure	lumpsum	280000

Query 21: Display projects where the achieved quantity in any single situation exceeds the expected quantity of the corresponding work package, showing the excess amount.

```
WITH AchieveDetails AS (
    SELECT
        a."Situation_id",
        a."WorkP_id",
        a."Achieved_quantity",
        wp."Expected_quantity",
        wp."WorkP_name",
        a."Achieved_quantity" - wp."Expected_quantity" AS Excess
    FROM "Achieve" a
    JOIN "WorkP" wp ON a."WorkP_id" = wp."WorkP_id"
    WHERE a."Achieved_quantity" > wp."Expected_quantity"
),
ProjectContext AS (
    SELECT
        p."Project_name",
        s."Comments" AS Situation_Description,
        ad."WorkP_name",
        ad."Expected_quantity",
        ad."Achieved_quantity",
        ad.Excess
    FROM AchieveDetails ad
    JOIN "Situation" s ON ad."Situation_id" = s."Situation_id"
    JOIN "Project" p ON s."Project_id" = p."Project_id"
)
SELECT * FROM ProjectContext
ORDER BY Excess DESC;
```

Query Result x					
All Rows Fetched: 2 in 0.011 seconds					
Project_name	SITUATION_DESCRIPTION	WorkP_name	Expected_quantity	Achieved_quantity	EXCESS
1 Sousse Swimming Pool	RC walls and pool floor	RC Walls Floor	420	600	180
2 Sousse Swimming Pool	Excavation and foundations	RC Walls Floor	420	500	80

Query 22: Display a full progress report per project and work package, showing expected quantity, total achieved, remaining quantity, completion percentage, and earned value.

```

WITH Progress AS (
    SELECT
        p."Project_name",
        wp."WorkP_name",
        wp."Measurment_unit",
        wp."Expected_quantity",
        wp."Unit_price",
        NVL(SUM(a."Achieved_quantity"), 0) AS Total_Achieved
    FROM "Project" p
    JOIN SD sd          ON p."SD_id"          = sd."SD_id"
    JOIN "WorkP" wp      ON wp."SD_id"        = sd."SD_id"
    LEFT JOIN "Situation" s ON s."Project_id"   = p."Project_id"
    LEFT JOIN "Achieve" a
        ON a."Situation_id" = s."Situation_id"
        AND a."WorkP_id"    = wp."WorkP_id"
    GROUP BY
        p."Project_name", wp."WorkP_name",
        wp."Measurment_unit", wp."Expected_quantity", wp."Unit_price"
)
SELECT
    "Project_name",
    "WorkP_name",
    "Measurment_unit",
    "Expected_quantity",
    Total_Achieved,
    "Expected_quantity" - Total_Achieved          AS Remaining_Quantity,
    ROUND(Total_Achieved
        / "Expected_quantity" * 100, 2)          AS Completion_Pct,
    ROUND(Total_Achieved * "Unit_price", 2)       AS Earned_Value
FROM Progress
ORDER BY "Project_name", Completion_Pct DESC;

```

Query Result x							
SQL All Rows Fetched: 40 in 0.024 seconds							
Project_name	WorkP_name	Measurment_unit	Expected_quantity	TOTAL_ACHIEVED	REMAINING_QUANTITY	COMPLETION_PCT	EARNED_VALUE
1 Ariana Tech Park	Earthworks	m3	900	1500	-600	166.67	69000
2 Ariana Tech Park	Cable Networks	lumpsum	1	1	0	100	72000
3 Ariana Tech Park	Reinforced Con.	m3	1200	1030	170	85.83	334750
4 Bizerte Complex	Slab Concrete	m3	560	1040	-480	185.71	312000
5 Bizerte Complex	Earthworks	m3	400	650	-250	162.5	28600
6 Bizerte Complex	Roof Cover	m2	900	900	0	100	85500
7 Carthage Annex B	Earthworks	m3	200	320	-120	160	14080
8 Carthage Annex B	Reinforced Con.	m3	350	530	-180	151.43	164300
9 Carthage Annex B	Masonry	m2	900	900	0	100	72000
10 Carthage Residency	Masonry	m2	2400	5400	-3000	225	459000
11 Carthage Residency	Reinforced Con.	m3	800	1500	-700	187.5	480000
12 Carthage Residency	Earthworks	m3	500	780	-280	156	35100
13 Carthage Residency	Facade Coating	m2	1800	2700	-900	150	102600
14 Hammamet Apartment	Superstructure	lumpsum	1	2	-1	200	560000
15 Hammamet Apartment	Infrastructure	lumpsum	1	2	-1	200	320000
16 Hammamet Apartment	Interior Walls	m2	1800	1800	0	100	63000
17 Hammamet Clinic	RC Structure	m2	3200	6200	-3000	193.75	1302000
18 Hammamet Clinic	Foundations	m3	750	1150	-400	153.33	339250
19 Hammamet Clinic	Medical Fluids	lumpsum	1	1	0	100	145000
20 Hammamet Clinic	Plasterboard	m2	4000	1000	3000	25	40000
21 Industrial Zone Exp.	Industrial Slab	m2	3500	6800	-3300	194.29	748000
22 Industrial Zone Exp.	Earthworks	m3	1200	1800	-600	150	82800
23 Industrial Zone Exp.	Steel Frame	tonne	80	120	-40	150	252000
24 Industrial Zone Exp.	Utility Networks	lumpsum	1	1	0	100	95000
25 Kairouan School	Shell Works	lumpsum	1	1	0	100	120000
26 Kairouan School	Finishing Works	lumpsum	1	0	1	0	0

Query 23: Generate a complete financial dashboard per project: total calculated cost, total earned value from achievements, budget consumption percentage, remaining budget, and a financial health status evaluated as OVER BUDGET, AT RISK, or ON TRACK.

```

WITH PlannedCost AS (
    SELECT
        p."Project_id",
        p."Project_name",
        p."Budget",
        SUM(wp."Unit_price" * wp."Expected_quantity") AS Total_Planned_Cost
    FROM "Project" p
    JOIN SD sd ON p."SD_id" = sd."SD_id"
    JOIN "WorkP" wp ON wp."SD_id" = sd."SD_id"
    GROUP BY p."Project_id", p."Project_name", p."Budget"
),
EarnedValue AS (
    SELECT
        p."Project_id",
        NVL(SUM(a."Achieved_quantity" * wp."Unit_price"), 0) AS Total_Earned
    FROM "Project" p
    JOIN SD sd ON p."SD_id" = sd."SD_id"
    JOIN "WorkP" wp ON wp."SD_id" = sd."SD_id"
    LEFT JOIN "Situation" s ON s."Project_id" = p."Project_id"
    LEFT JOIN "Achieve" a ON a."Situation_id" = s."Situation_id"
        AND a."WorkP_id" = wp."WorkP_id"
    GROUP BY p."Project_id"
),
Dashboard AS (
    SELECT
        pc."Project_name",
        pc."Budget",
        pc.Total_Planned_Cost,
        ev.Total_Earned,
        ROUND(ev.Total_Earned / pc."Budget" * 100, 2)
            AS Budget_Consumption_Pct,
        pc."Budget" - ev.Total_Earned AS Remaining_Budget
    FROM PlannedCost pc
    JOIN EarnedValue ev ON pc."Project_id" = ev."Project_id"
)
SELECT
    "Project_name",
    "Budget",
    Total_Planned_Cost,
    Total_Earned,
    Budget_Consumption_Pct,
    Remaining_Budget,
    CASE
        WHEN Budget_Consumption_Pct > 90 THEN 'OVER BUDGET'
        WHEN Budget_Consumption_Pct > 60 THEN 'AT RISK'
        ELSE 'ON TRACK'
    END AS Financial_Status
FROM Dashboard
ORDER BY Budget_Consumption_Pct DESC;

```

Query Result x						
SQL All Rows Fetched: 12 in 0.02 seconds						
Project_name	Budget	TOTAL_PLANNED_COST	TOTAL_EARNED	BUDGET_CONSUMPTION_PCT	REMAINING_BUDGET	FINANCIAL_STATUS
1 Hammamet Clinic	1100000	1198250	1826250	166.02	-726250	OVER BUDGET
2 Hammamet Apartment	630000	503000	943000	149.68	-313000	OVER BUDGET
3 Industrial Zone Exp.	850000	703200	1177800	138.56	-327800	OVER BUDGET
4 Sousse Swimming Pool	220000	118200	258000	117.27	-38000	OVER BUDGET
5 Nabeul North Villa	480000	233400	425400	88.63	54600	AT RISK
6 Carthage Residency	1250000	550900	1076700	86.14	173300	AT RISK
7 Sousse Beach Hotel	1850000	731300	1243300	67.21	606700	AT RISK
8 Carthage Annex B	390000	189300	250380	64.2	139620	AT RISK
9 Ariana Tech Park	780000	503400	475750	60.99	304250	AT RISK
10 Tunis Lake Tower	1950000	1741500	937500	48.08	1012500	ON TRACK
11 Bizerte Complex	975000	271100	426100	43.7	548900	ON TRACK
12 Kairouan School	310000	220000	120000	38.71	190000	ON TRACK

Table 7: SQL query formulation and execution results

Conclusion

This chapter detailed the formulation of SQL queries and their execution within the Oracle environment. In the following chapter, we will describe the implementation and development of our web application.

Chapter 5. Web Application Implementation

Introduction

The last step of the development process concerns the implementation of the application based on the chosen technologies. It constitutes the final part of this report and aims to present the completed work, demonstrating that the designed system is operational and capable of satisfying the construction site monitoring needs defined at the beginning of the project.

1. Overall solution architecture

ConstructManager adopts a classic three-tier web architecture, which separates concerns into three distinct layers and ensures maintainability, scalability, and clarity of the codebase as it is illustrated in Table 8.

Tier	Role	Technologies Used
Presentation Tier	Displays the user interface in the browser; handles user input and navigation.	HTML5, CSS3, JavaScript, Font Awesome icons
Application Tier	Contains all business logic, routing, and data processing.	PHP 8 (MVC pattern), Apache via XAMPP
Data Tier	Persists all application data in a relational database.	MySQL via XAMPP / phpMyAdmin

Table 8 : Three-Tier Architecture of ConstructManager

This separation into three levels, illustrated in Figure 10, simplifies software installation procedures, information sharing between components, and enables the reuse of application modules.

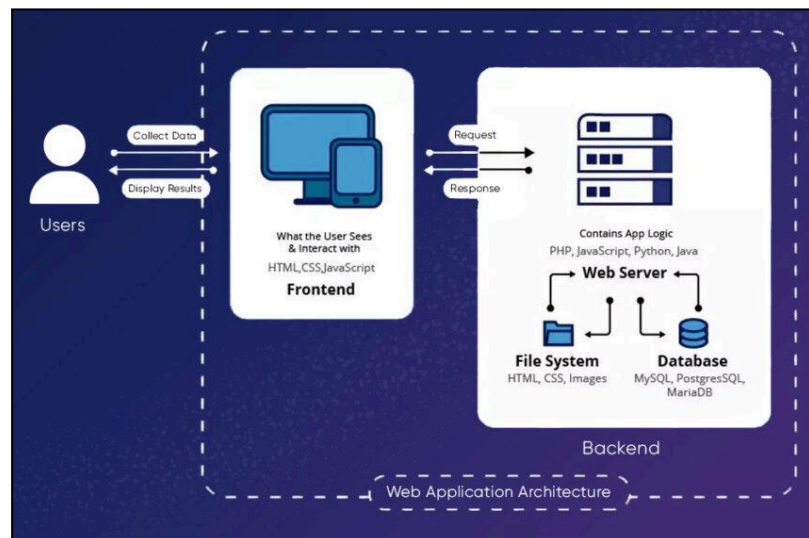


Figure 10: Overall architecture of the web application

2. DataBase Connection

The database was created using phpMyAdmin containing all seven tables defined in the physical data model: Client, Project, SD, Work_P, Situation, Achieve, and Document. (Figure11).

Table	Action	Rows	Type	Collation	Size	Overhead
☐ achieve	★ Browse Structure Search Insert Empty Drop	85	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ client	★ Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ document	★ Browse Structure Search Insert Empty Drop	31	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ project	★ Browse Structure Search Insert Empty Drop	15	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ sd	★ Browse Structure Search Insert Empty Drop	15	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ situation	★ Browse Structure Search Insert Empty Drop	37	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ work_p	★ Browse Structure Search Insert Empty Drop	49	InnoDB	utf8mb4_general_ci	32.0 KiB	-

Figure 11 : Database Tables Overview

Once the database was created, a connection was established using PHP PDO to allow the application to store information submitted by users through the web interface and to display data already inserted in the database. The connection is centralized in a single file config/database.php, ensuring it is reused across all controllers without redundancy.(Figure12)

```

config > database.php
1  <?php
2  class Database {
3      private $host = "localhost";
4      private $db_name = "construction_management";
5      private $username = "root";
6      private $password = "";
7      public $conn;
8
9      public function getConnection() {
10         $this->conn = null;
11         try {
12             $this->conn = new PDO("mysql:host=" . $this->host . ";dbname=" . $this->db_name,
13                                 $this->username, $this->password);
14             $this->conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
15             $this->conn->exec("set names utf8");
16         } catch(PDOException $exception) {
17             echo "Connection error: " . $exception->getMessage();
18         }
19         return $this->conn;
20     }
21 }
22 ?>

```

Figure 12: The database connection code

3. Application Realization

We will present in this section an overview of the realized web application, illustrating its main interfaces and functionalities.

3.1. Login Page

Access to the Dashboard is protected by an authentication system. Figure 13 illustrates the login page, which requires a username and password before granting access to the system. All pages of the application are protected by a session check that redirects unauthenticated users to this page.

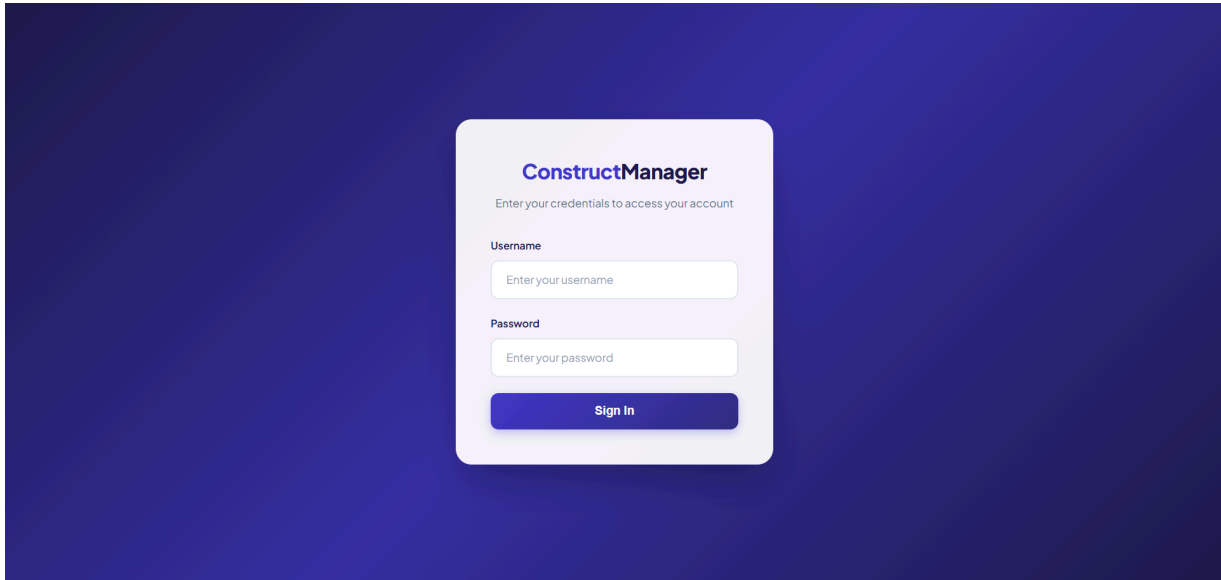


Figure 13: Login page

3.2. Dashboard

Figure 14 illustrates the main dashboard, which serves as the home page of the application after login. It presents a global overview of the construction firm's activity through four summary cards showing the total number of projects, clients, active situations, and stored documents.

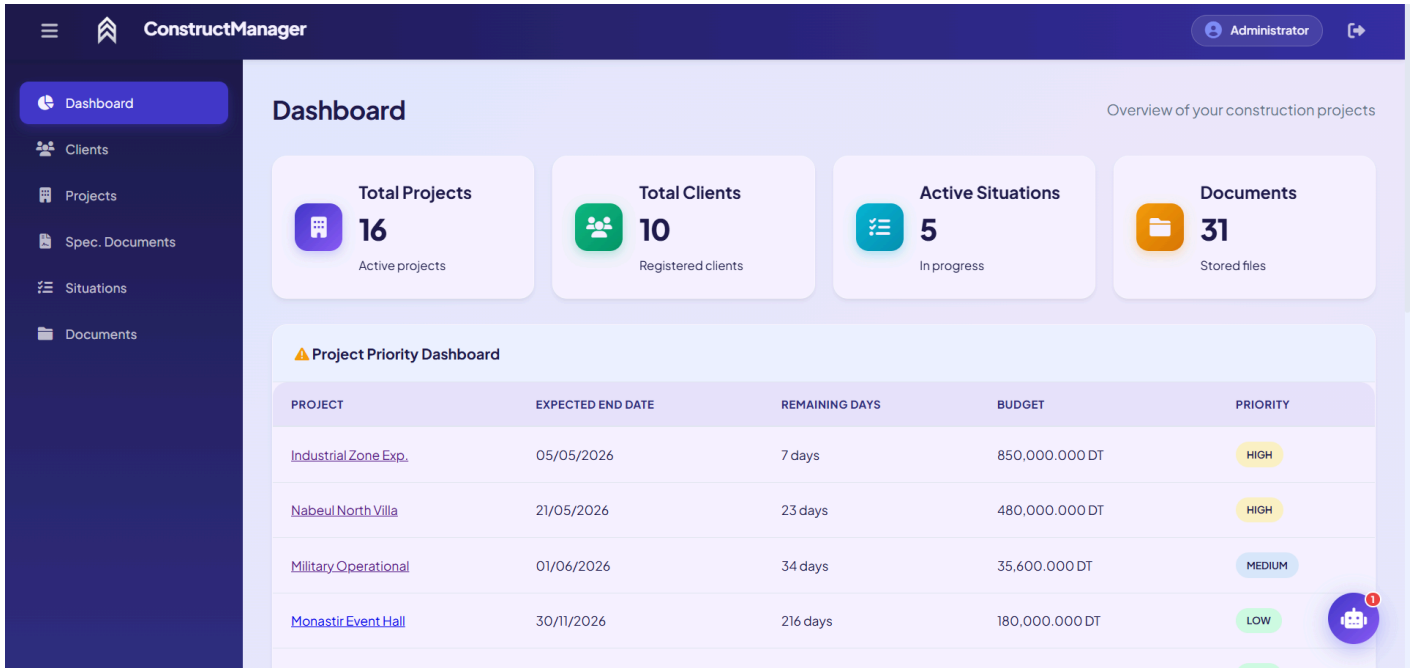


Figure 14: Main dashboard — summary cards

Below the summary cards, the dashboard displays a **Project Priority Dashboard** table listing all projects with their expected end date, remaining days, budget, and an automatically calculated priority badge. This priority is computed dynamically from the current date.

Project Priority Dashboard				
PROJECT	EXPECTED END DATE	REMAINING DAYS	BUDGET	PRIORITY
Industrial Zone Exp.	05/05/2026	7 days	850,000.000 DT	HIGH
Nabeul North Villa	21/05/2026	23 days	480,000.000 DT	HIGH
Military Operational	01/06/2026	34 days	35,600.000 DT	MEDIUM
Sfax Warehouse	31/12/2026	247 days	410,000.000 DT	LOW
Monastir Parking	19/01/2027	266 days	155,000.000 DT	LOW
Kairouan School	30/06/2027	428 days	310,000.000 DT	LOW
Tunis Lake Tower	31/01/2028	643 days	1,950,000.000 DT	LOW

Figure 15: Project Priority Dashboard with colored priority badges

The dashboard also includes a Recent Projects panel showing the latest projects with their progress bars, and a Recent Situations panel showing the most recently recorded progress statements.

Recent Projects View all →				Recent Situations View all →		
PROJECT	CLIENT	BUDGET	PROGRESS	PROJECT	COMMENT	PERIOD
Hotel RosaBeach	Riadh Jebali	14,500.000 DT	24%	Hotel RosaBeach		28/04/2026 - 15/04/2026
Kairouan School	Nadia Bouaziz	310,000.000 DT	33.33%	Kairouan School	Shell works site kickoff	15/03/2026 - 15/04/2026
Monastir Event Hall	Sonia Khelifi	180,000.000 DT	33.33%	Monastir Parking	RC shear walls level 1	01/03/2026 - 31/03/2026
Tunis Lake Tower	Riadh Jebali	1,950,000.000 DT	53.7%	Tunis Lake Tower	Raft foundation started	01/03/2026 - 31/03/2026
Tunis Lake Tower		1,950,000.000 DT	0%	Tunis Lake Tower	Bored piles 80 units done	01/02/2026 - 28/02/2026

Figure 16: Recent projects and recent situations panels

3.3. Projects Page

Figure 17 illustrates the projects list page, which displays all construction projects managed by the firm. Each row shows the project name, associated client, budget, start and end dates, overall completion progress bar, and corresponding actions. A search bar allows filtering projects by name or client name in real time.

ConstructManager Administrator						
Project Management + New Project						
Track the progress of all your construction projects						
Search...						
ID	PROJECT	CLIENT	BUDGET	DATES	PROGRESS	ACTIONS
#8	Kairouan School SD vv1.1	Nadia Bouaziz	310,000.000 DT	15/03/2026 30/06/2027	33.33%	   
#15	Monastir Event Hall SD vv1.0	Sonia Khelifi	180,000.000 DT	01/02/2026 07/04/2026	33.33%	   
#7	Tunis Lake Tower SD vv1.0	Riadh Jebali	1,950,000.000 DT	01/02/2026 31/01/2028	53.7%	   
#9	Monastir Parking SD vv1.0	Hassen Dridi	155,000.000 DT	20/01/2026 19/01/2027	83.33%	   
#6	Sfax Warehouse SD vv2.0	Sonia Khelifi	410,000.000 DT	01/09/2025 31/12/2026	86.67%	   

Figure 17: Projects list page

3.4. Project Detail Page

Figure 18 illustrates the project detail page, which is the central and most important interface of the application. It presents all information related to a single project in one place. At the top, a detail card shows the project name, client full name and address, specification document version, and a circular SVG progress gauge showing the overall completion percentage of the project.

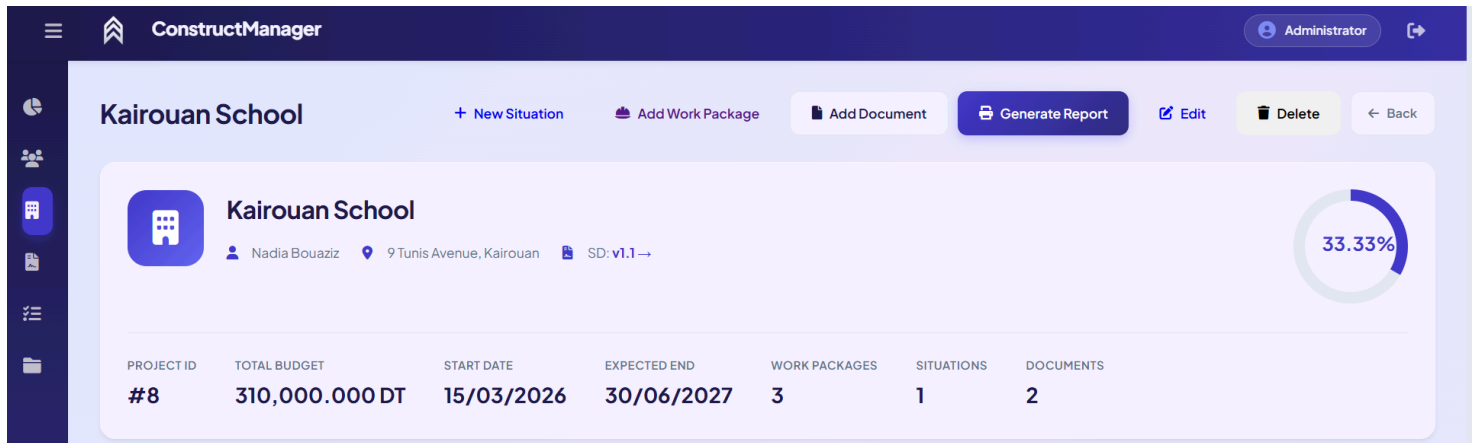


Figure 18: Project detail page

Below the header, a strip of statistics shows the project ID, total budget, start date, expected end date, number of work packages, number of situations, and number of documents.

The page contains three tabs — Works, Situations, and Documents — allowing navigation between the different aspects of the project. The Works tab illustrated in Figure 19 displays all work packages belonging to the project's specification document, with their expected quantity, unit, unit price, total achieved quantity, a color-coded completion progress bar (red below 30%, orange between 30% and 75%, green above 75%), a color-coded remaining quantity, and the total earned value per work package. A grand total row summarizes the overall earned value at the bottom of the table.

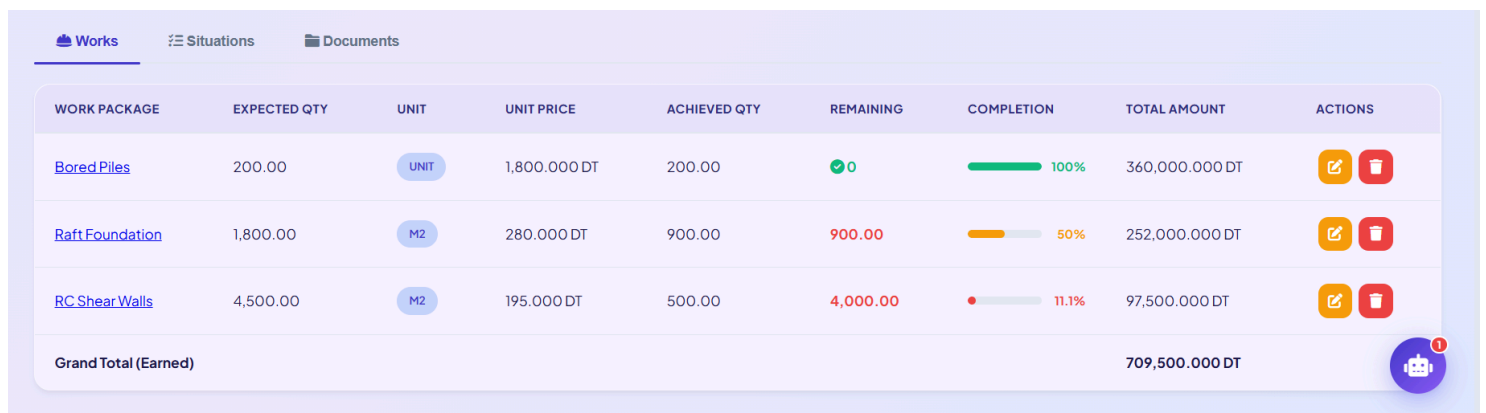
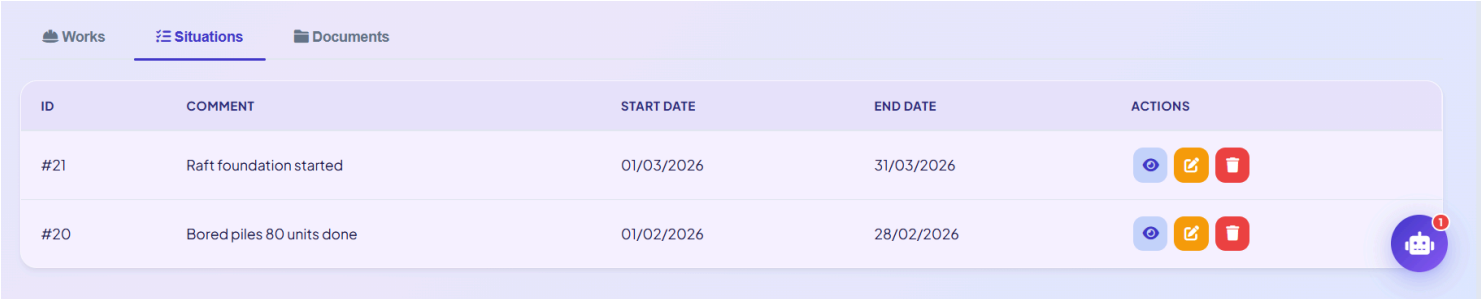


Figure 19: Project detail page — Works tab with color-coded progress bars

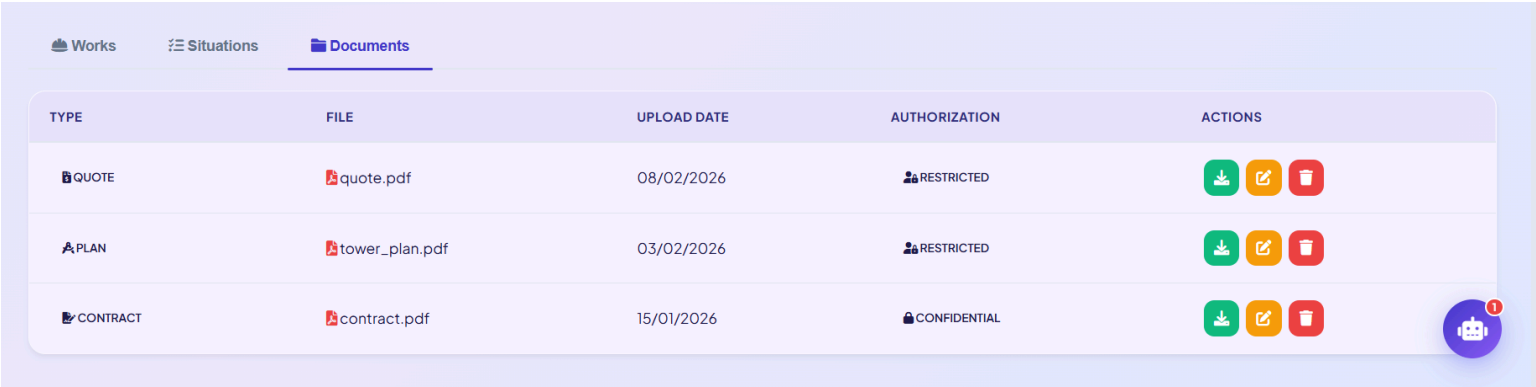
The Situations tab illustrated in Figure 20 lists all progress statements recorded for the project, with their dates, comments, and action buttons to view, edit, or delete each situation.



ID	COMMENT	START DATE	END DATE	ACTIONS
#21	Raft foundation started	01/03/2026	31/03/2026	  
#20	Bored piles 80 units done	01/02/2026	28/02/2026	  

Figure 20: Project detail page — Situations tab

The Documents tab lists all contractual documents associated with the project, showing their type, upload date, authorization level, and download,edit and delete buttons.



TYPE	FILE	UPLOAD DATE	AUTHORIZATION	ACTIONS
QUOTE	 quote.pdf	08/02/2026	 RESTRICTED	  
PLAN	 tower_plan.pdf	03/02/2026	 RESTRICTED	  
CONTRACT	 contract.pdf	15/01/2026	 CONFIDENTIAL	  

Figure 21: Project detail page — Documents tab

Figure 22 illustrates the financial health card displayed above the tabs on each project detail page. This card is generated from a dedicated SQL query and presents the total planned cost of all work packages, the total earned value based on achieved quantities, a color-coded budget consumption progress bar, the remaining budget, and a financial status badge evaluated automatically as ON TRACK, AT RISK, or OVER BUDGET.



Figure 22: Financial health card with budget consumption bar and status badge

This feature provides construction managers with an immediate financial overview of each project without needing to perform manual calculations.

Figure 23 illustrates the auto-generated printable report available for each project. By clicking the "**Generate Report**" button on the project detail page, a new browser tab opens displaying a complete standalone HTML report containing the project information, financial health summary, a full work package progress table, and a situations history table. A Print button at the top of the report allows the user to export or print the document directly.

PROJECT INFORMATION

PROJECT ID #9	CLIENT Hassen Dridi	SD VERSION v1.0
START DATE 20/01/2026	EXPECTED END 19/01/2027	OVERALL PROGRESS 83.33%

FINANCIAL SUMMARY

BUDGET 155,000.000 DT	PLANNED COST 646,000.000 DT	EARNED VALUE 535,000.000 DT	REMAINING -380,000.000 DT
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WORK PACKAGES

#	Work Package	Unit	Expected Qty	Achieved Qty	Remaining	Unit Price (DT)	Earned (DT)	Progress
28	Bored Piles	unit	120.00	120.00	✓ 0	1,800.000	216,000.000	100%
29	Parking Raft	m2	800.00	800.00	✓ 0	260.000	208,000.000	100%
30	RC Shear Walls	m2	1,200.00	600.00	600.00	185.000	111,000.000	50%
TOTAL							535,000.000 DT	

SITUATIONS

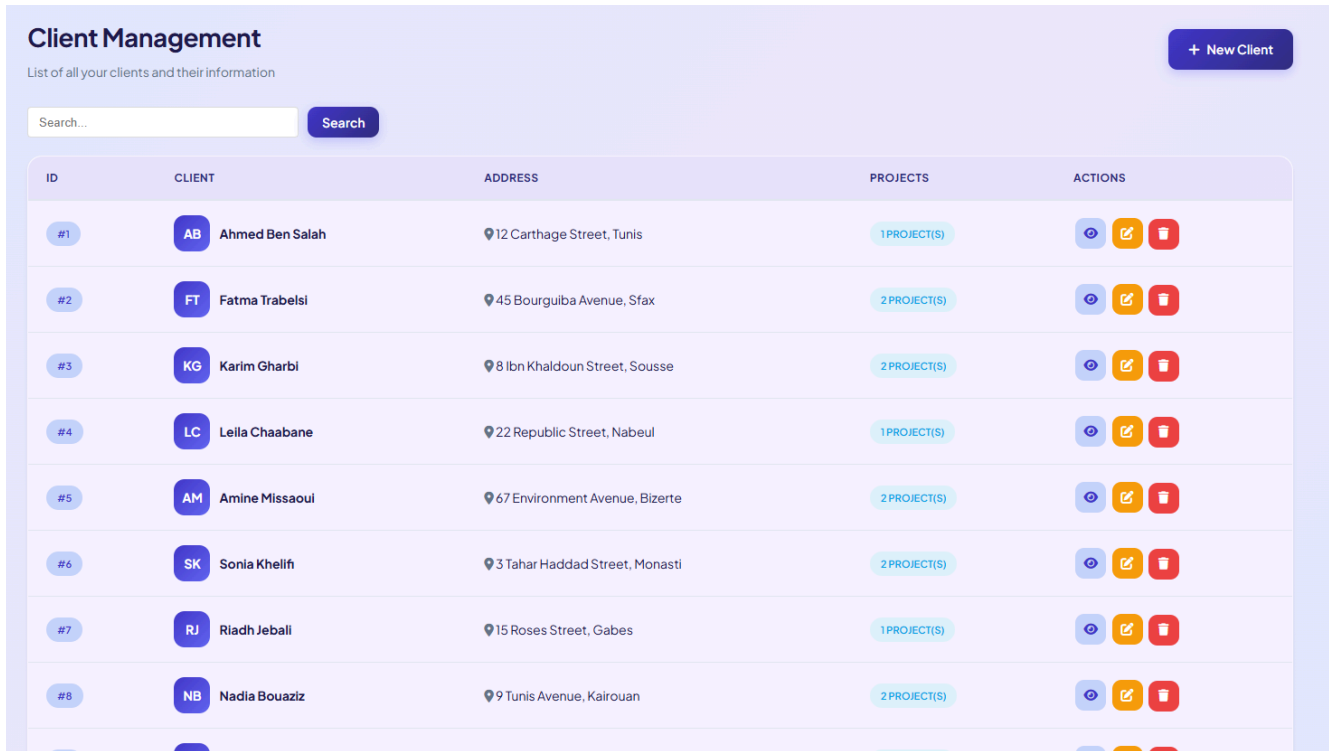
ID	Comments	Start Date	End Date	Duration (days)
#24	RC shear walls level 1	01/03/2026	31/03/2026	29 days
#23	Piles and raft slab	20/01/2026	20/02/2026	31 days

Figure 23: Auto-generated printable situation report

This feature demonstrates that the system produces actionable output that construction managers can share or archive, going beyond simple data display.

3.5. Clients Page

Figure 24 illustrates the clients page, which lists all registered clients with their full name, address, and number of associated projects. Clicking on a client reveals their details and the list of projects they have commissioned.



The screenshot shows a 'Client Management' interface. At the top, there's a header with the title 'Client Management' and a subtitle 'List of all your clients and their information'. A '+ New Client' button is in the top right. Below the header is a search bar with a 'Search' button. The main content is a table with columns: ID, CLIENT, ADDRESS, PROJECTS, and ACTIONS. The table lists 8 clients with their respective details and actions (view, edit, delete).

ID	CLIENT	ADDRESS	PROJECTS	ACTIONS
#1	AB Ahmed Ben Salah	12 Carthage Street, Tunis	1 PROJECT(S)	
#2	FT Fatma Trabelsi	45 Bourguiba Avenue, Sfax	2 PROJECT(S)	
#3	KG Karim Gharbi	8 Ibn Khaldoun Street, Sousse	2 PROJECT(S)	
#4	LC Leila Chaabane	22 Republic Street, Nabeul	1 PROJECT(S)	
#5	AM Amine Missaoui	67 Environment Avenue, Bizerte	2 PROJECT(S)	
#6	SK Sonia Khelif	3 Tahar Haddad Street, Monasti	2 PROJECT(S)	
#7	RJ Riadh Jebali	15 Roses Street, Gabes	1 PROJECT(S)	
#8	NB Nadia Bouaziz	9 Tunis Avenue, Kairouan	2 PROJECT(S)	

Figure 24: Clients list page

3.6. Add and Edit Forms

Figure 25 illustrates an example of a data entry form, here the project creation form. The form allows the user to enter the project name, select a client from a dropdown, define the budget, start date, and expected end date. Similar forms exist for all entities in the system — clients, work packages, situations, and documents — ensuring full data management capabilities.

Figure 25: Project creation form

3.7. Integrated AI Assistant Chatbot

Figure 26 illustrates the integrated AI assistant chatbot available throughout the entire application. A chat icon is permanently displayed in the bottom-right corner of every page, allowing users to interact with an intelligent assistant at any moment without leaving their current interface. The chatbot is powered by a local **Ollama** language model, meaning it runs entirely on the local machine without requiring any external API or internet connection, which guarantees data confidentiality — a critical requirement for a construction firm managing sensitive contractual and financial information.

The assistant is capable of answering questions related to construction site management, helping users understand the system, and providing guidance on how to interpret the data displayed on the platform.

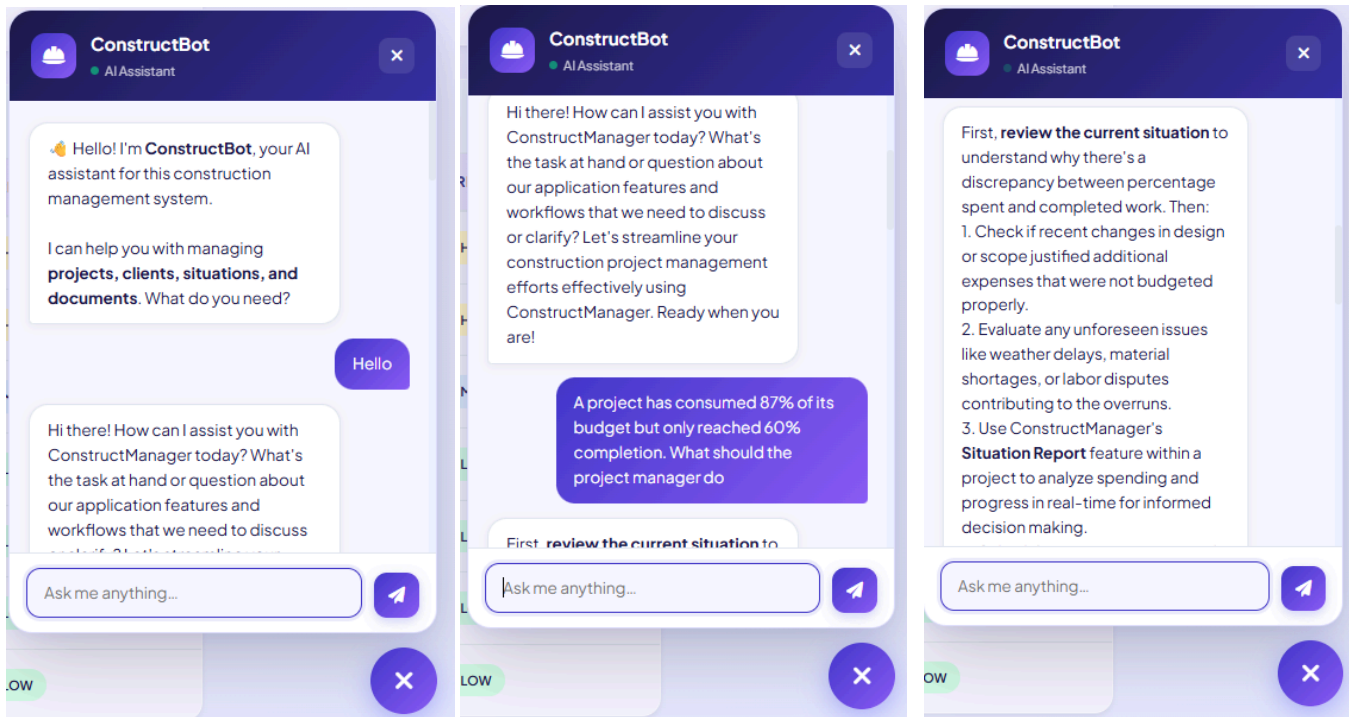


Figure 26: AI assistant chatbot

Conclusion

This chapter described the architecture and the main interfaces of the Construction Site Monitoring System web application. The developed solution successfully bridges the gap between the relational database designed in the previous chapters and the daily operational needs of construction managers. Through an intuitive PHP interface backed by a structured MySQL database, users can manage clients, projects, work packages, progress situations, and documents in a centralized environment. The addition of the financial health card, color-coded progress indicators, the auto-generated printable report and the chatbot demonstrates that the system goes beyond basic data management to provide genuine decision-support tools for construction site monitoring.

General Conclusion

In the framework of our Year-End Project, we set out to bridge the gap between theoretical engineering concepts and practical implementation. This project allowed us to navigate the essential constraints of professional development while preparing us for our future graduation projects.

We have designed and developed a dedicated **Construction Site Tracking** platform, providing a centralized tool for monitoring project progress, financial budgets, and document management. This document has detailed our entire journey, from the preliminary requirements analysis to the physical deployment of the database and web interface.

This work has been of significant importance for several reasons. Firstly, it allowed us to deepen our technical expertise in database normalization and SQL optimization. Secondly, it provided a realistic perspective on the challenges inherent in project management, such as effort distribution and time constraints. Furthermore, working as a team has been an enriching experience, strengthening our communication skills and our ability to collaborate effectively toward a common technical goal.

However, our work does not end here. We recognize that several improvements could further elevate the platform's value. Future perspectives include:

- **Implementing role-based access control**, introducing multiple user profiles such as project manager, site engineer, and client, each with tailored permissions and dedicated dashboards, ensuring that sensitive financial and contractual data remains accessible only to authorized personnel.
- **Developing a dedicated mobile application** for Android and iOS platforms, enabling field engineers to update work package progress, attach photos, and submit situation reports directly from construction sites in real time.
- **Integrating automated financial alerts**, where the system proactively notifies responsible parties by email or in-app notification when a project's budget consumption crosses predefined thresholds, allowing early intervention before cost overruns occur.
- **Connecting to external accounting and ERP systems** via a REST API layer, allowing ConstructManager to synchronize invoices, supplier payments, and payroll data with existing enterprise tools used by construction firms.

Ultimately, this project has been a vital milestone in our engineering education, proving that a well-structured data model is the backbone of any successful industrial management solution.

Bibliography

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Glossary / Acronyms

BTP:	Bâtiment et Travaux Publics (Construction and Civil Engineering)
SQL:	Structured Query Language
DB / DBMS:	Database / Database Management System
CDM:	Conceptual Data Model
LDM:	Logical Data Model
PDM:	Physical Data Model
SD:	Specification Document (Cahier des Charges)
WorkP:	Work Package
PHP:	Hypertext Preprocessor
XAMPP:	Cross-platform Apache MySQL PHP Perl
ENSIT:	École Nationale Supérieure d'Ingénieurs de Tunis

Appendices

The database creation script:

```
/*=====*/
/* DBMS name:      ORACLE Version 11g                               */
/* Created on:     4/1/2026 10:24:44 PM                             */
/*=====*/

alter table "Achieve"
  drop constraint FK_ACHIEVE_ACHIEVE_WORKP;

alter table "Achieve"
  drop constraint FK_ACHIEVE_ACHIEVE2_SITUATIO;

alter table "Document"
  drop constraint FK_DOCUMENT_CENTRALIZ_PROJECT;

alter table "Project"
  drop constraint FK_PROJECT_ORDER_CLIENT;

alter table "Project"
  drop constraint FK_PROJECT_STRUCTURE_SD;

alter table "Situation"
  drop constraint FK_SITUATIO_ESTABLISH_PROJECT;

alter table "WorkP"
  drop constraint FK_WORKP_CONTAIN_SD;

drop table "Achieve" cascade constraints;

drop table "Client" cascade constraints;

drop table "Document" cascade constraints;

drop table "Project" cascade constraints;

drop table SD cascade constraints;

drop table "Situation" cascade constraints;

drop table "WorkP" cascade constraints;

/*=====*/
/* Table: "Achieve"                                               */
/*=====*/
create table "Achieve"
(
  "Situation_id"      NUMBER(8)          not null,
  "WorkP_id"          NUMBER(8)          not null,
  "Achieved_quantity" NUMBER(8),
  constraint PK_ACHIEVE primary key ("Situation_id", "WorkP_id")
);

/*=====*/
/* Table: "Client"                                               */
/*=====*/
```



```

/*=====*/
create table "Client"
(
    "Client_id"          NUMBER(8)          not null,
    "First_name"         VARCHAR2(20),
    "Last_name"          VARCHAR2(20),
    "Address"            VARCHAR2(30),
    constraint PK_CLIENT primary key ("Client_id")
);

/*=====*/
/* Table: "Document" */
/*=====*/
create table "Document"
(
    "Document_id"        NUMBER(8)          not null,
    "Project_id"         NUMBER(8)          not null,
    "Document_type"      VARCHAR2(20),
    "Document_url"       VARCHAR2(255),
    "Upload_date"        DATE,
    "Authorization_level" VARCHAR2(20),
    constraint PK_DOCUMENT primary key ("Document_id")
);

/*=====*/
/* Table: "Project" */
/*=====*/
create table "Project"
(
    "Project_id"         NUMBER(8)          not null,
    "Client_id"          NUMBER(8)          not null,
    "SD_id"              NUMBER(8),
    "Project_name"       VARCHAR2(20),
    "Budget"             NUMBER(10,3),
    "Project_start_date" DATE,
    "Project_expected_end_date" DATE,
    constraint PK_PROJECT primary key ("Project_id")
);

/*=====*/
/* Table: SD */
/*=====*/
create table SD
(
    "SD_id"              NUMBER(8)          not null,
    "date_creation"      DATE,
    "Version"            VARCHAR2(20),
    constraint PK_SD primary key ("SD_id")
);

/*=====*/
/* Table: "Situation" */
/*=====*/
create table "Situation"
(
    "Situation_id"       NUMBER(8)          not null,
    "Project_id"         NUMBER(8)          not null,
    "Comments"           VARCHAR2(100),
    "Start_date"         DATE,
    "End_date"           DATE,
    constraint PK_SITUATION primary key ("Situation_id")
);

```

```

/*=====*/
/* Table: "WorkP" */
/*=====*/
create table "WorkP"
(
    "WorkP_id"          NUMBER(8)          not null,
    "SD_id"             NUMBER(8)          not null,
    "WorkP_name"        VARCHAR2(20),
    "Expected_quantity" NUMBER(8),
    "Measurment_unit"   VARCHAR2(10),
    "Unit_price"        NUMBER(10,3),
    constraint PK_WORKP primary key ("WorkP_id")
);

alter table "Achieve"
    add constraint FK_ACHIEVE_ACHIEVE_WORKP foreign key ("WorkP_id")
        references "WorkP" ("WorkP_id");

alter table "Achieve"
    add constraint FK_ACHIEVE_ACHIEVE2_SITUATIO foreign key ("Situation_id")
        references "Situation" ("Situation_id");

alter table "Document"
    add constraint FK_DOCUMENT_CENTRALIZ_PROJECT foreign key ("Project_id")
        references "Project" ("Project_id");

alter table "Project"
    add constraint FK_PROJECT_ORDER_CLIENT foreign key ("Client_id")
        references "Client" ("Client_id");

alter table "Project"
    add constraint FK_PROJECT_STRUCTURE_SD foreign key ("SD_id")
        references SD ("SD_id");

alter table "Situation"
    add constraint FK_SITUATIO_ESTABLISH_PROJECT foreign key ("Project_id")
        references "Project" ("Project_id");

alter table "WorkP"
    add constraint FK_WORKP_CONTAIN_SD foreign key ("SD_id")
        references SD ("SD_id");

```

Abstract

This Year-End Project presents the design and implementation of a Construction Site Monitoring System for a civil engineering firm. The system provides a centralized relational database built on Oracle 11g, modeled through Conceptual, Logical, and Physical Data Models. It manages clients, projects, specification documents, work packages, progress situations, and contractual documents. Twenty-three SQL queries were developed and executed to meet the functional requirements of the system. A web application was additionally developed using PHP to provide an interactive interface for construction managers. This project strengthens our practical skills in database design, SQL optimization, and web development.

Keywords: Construction site monitoring, relational database, Oracle 11g, SQL, PHP, work packages.

Résumé

Ce Projet de Fin d'Année présente la conception et la mise en œuvre d'un Système de Suivi de Chantier pour une entreprise de génie civil. Le système repose sur une base de données relationnelle centralisée développée sous Oracle 11g, modélisée à travers les modèles conceptuel, logique et physique des données. Il permet la gestion des clients, des projets, des cahiers des charges, des postes de travaux, des situations d'avancement et des documents contractuels. Vingt-trois requêtes SQL ont été formulées et exécutées pour répondre aux besoins fonctionnels du système. Une application web développée en PHP offre une interface interactive aux responsables de chantier. Ce projet renforce nos compétences pratiques en conception de bases de données, optimisation SQL et développement web.

Mots clés : Suivi de chantier, base de données relationnelle, Oracle 11g, SQL, PHP, postes de travaux,

المخلص

يقدم هذا المشروع السنوي تصميم وتطوير نظام معلوماتي لمتابعة ورش البناء لدى شركة هندسة مدنية. يعتمد النظام على قاعدة بيانات علائقية مركزية مبنية على Oracle 11g، تم نمذجتها عبر النماذج المفاهيمية والمنطقية والمادية للبيانات. يتيح النظام إدارة العملاء والمشاريع ووثائق المواصفات وحزم الأعمال ووضعيات التقدم والوثائق التعاقدية. تم تطوير ثلاثة وعشرين استعلاماً بلغة SQL للإجابة عن المتطلبات الوظيفية للنظام. كما تم تطوير تطبيق ويب باستخدام PHP يوفر واجهة تفاعلية لمديري المشاريع.

الكلمات المفتاحية: متابعة ورش البناء، قاعدة البيانات العلائقية، PHP، SQL، Oracle 11g، حزم الأعمال،